

GAIRA Raman Reference Atlas

v0.1 — Component Audit and Biochemical Structure Analysis

Atlas	NMF k=24 (FROZEN, fingerprint 09ed804a4083...)
Reference corpus	375 Raman spectra · 167 pure analytes
Mean component stability	0.812
Mean class purity	0.347
Interpretive confidence	13 high · 11 moderate · 0 low
Recommended grouping	k=14 (interpretive overlay, NOT frozen)
BSV recommendation	Option C — hierarchical (24 canonical → themes)
Status	Audit only — nothing retrained, nothing frozen here

PRINCIPAL FINDING

The atlas has learned molecular CLASS, not molecular species. Lipids, carbohydrates and proteins each cohere against a random-analyte null; amino acids, purines, cofactors and organic acids do not. Components are stable (0.81) but only modestly pure (0.35), so the 24 components should stay canonical as the NUMERICAL layer while biochemical meaning is carried by a revisable, versioned theme overlay.

Executive summary

Frozen NMF k=24 · 167 analytes · audit only

What this audit asked

The Raman reference atlas (NMF, k=24) was already built and frozen. This study asks one question: what biochemical structure has it actually learned? Nothing was retrained; the atlas fingerprint was verified unchanged before and after the audit.

What the atlas has learned

Lipid chemistry dominates (triacylglycerols, fatty acids, sterols cohere with $p < 0.05$ against a random-analyte null), followed by carbohydrates (mono- and polysaccharides) and proteins. Nucleic-acid bases appear as strongly ENRICHED but small components (purine enrichment 10x, pyrimidine 14x). Amino acids, cofactors and organic acids do NOT form coherent groups at analyte level.

Components are stable but only moderately pure

Mean bootstrap component stability is 0.812 — the components reproduce well under analyte resampling. But mean class purity is only 0.35: most components are chemically ENRICHED rather than chemically pure. Enrichment reaches 14x (pyrimidine) and 10x (purine) even where purity is modest, which is the signature of a real but small chemical theme sitting inside a broader mixture. 13 of 24 components reach high interpretive confidence; none are uninterpretable.

Higher-order groups exist but are weak geometrically

Grouping is highly reproducible under analyte bootstrap (0.969) yet silhouette is only 0.095 across every k tested. The components therefore do NOT fall into well-separated natural clusters — grouping is a useful interpretive convenience, not a discovered partition. k=14 ranks best on the composite of reproducibility, chemical coherence, silhouette and interpretability, but the margin over neighbouring k is small and the count should be treated as provisional.

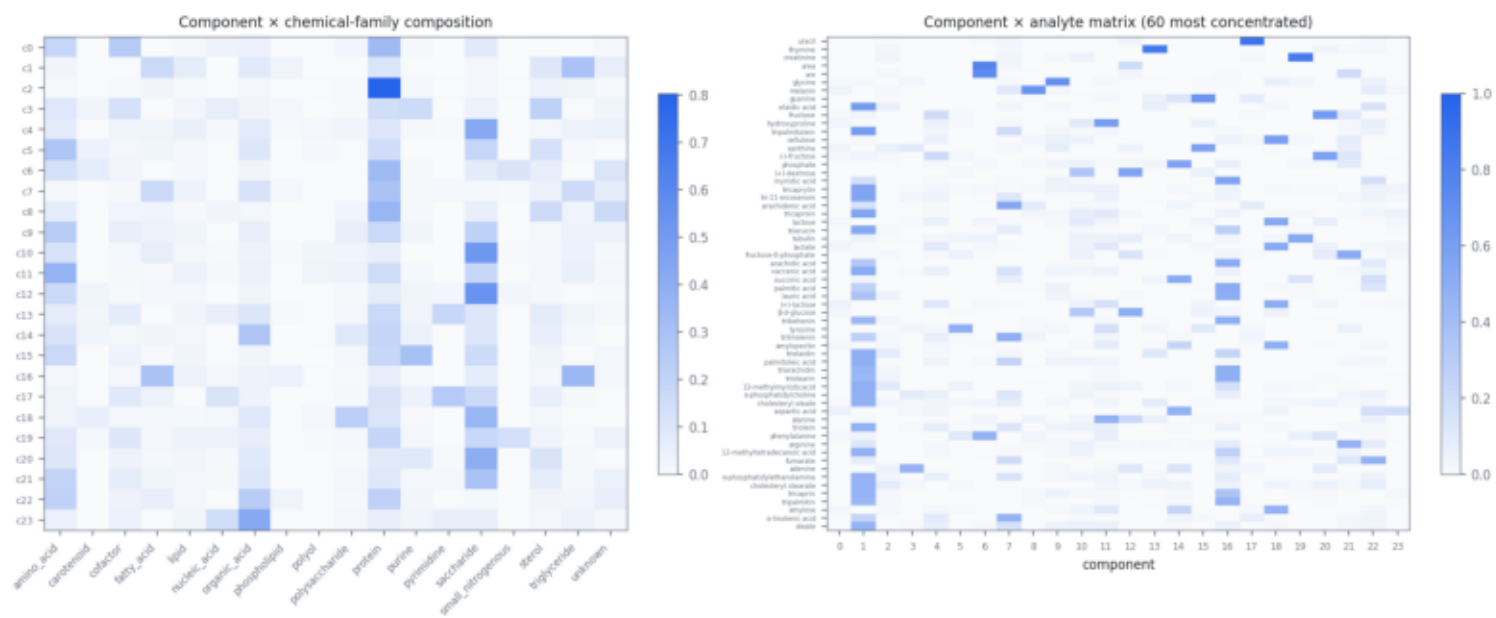
MSS: class-level yes, species-level no

Median signature uniqueness is 0.053, which looks alarming until the confusions are examined: 88% of low-uniqueness cases are duplicate names, homologous series (the saturated triacylglycerols differ only in acyl-chain length), or same-class chemistry. That is correct fingerprint-region spectroscopy, not atlas failure. Only 16 genuinely cross-chemistry confusions remain.

Recommendation

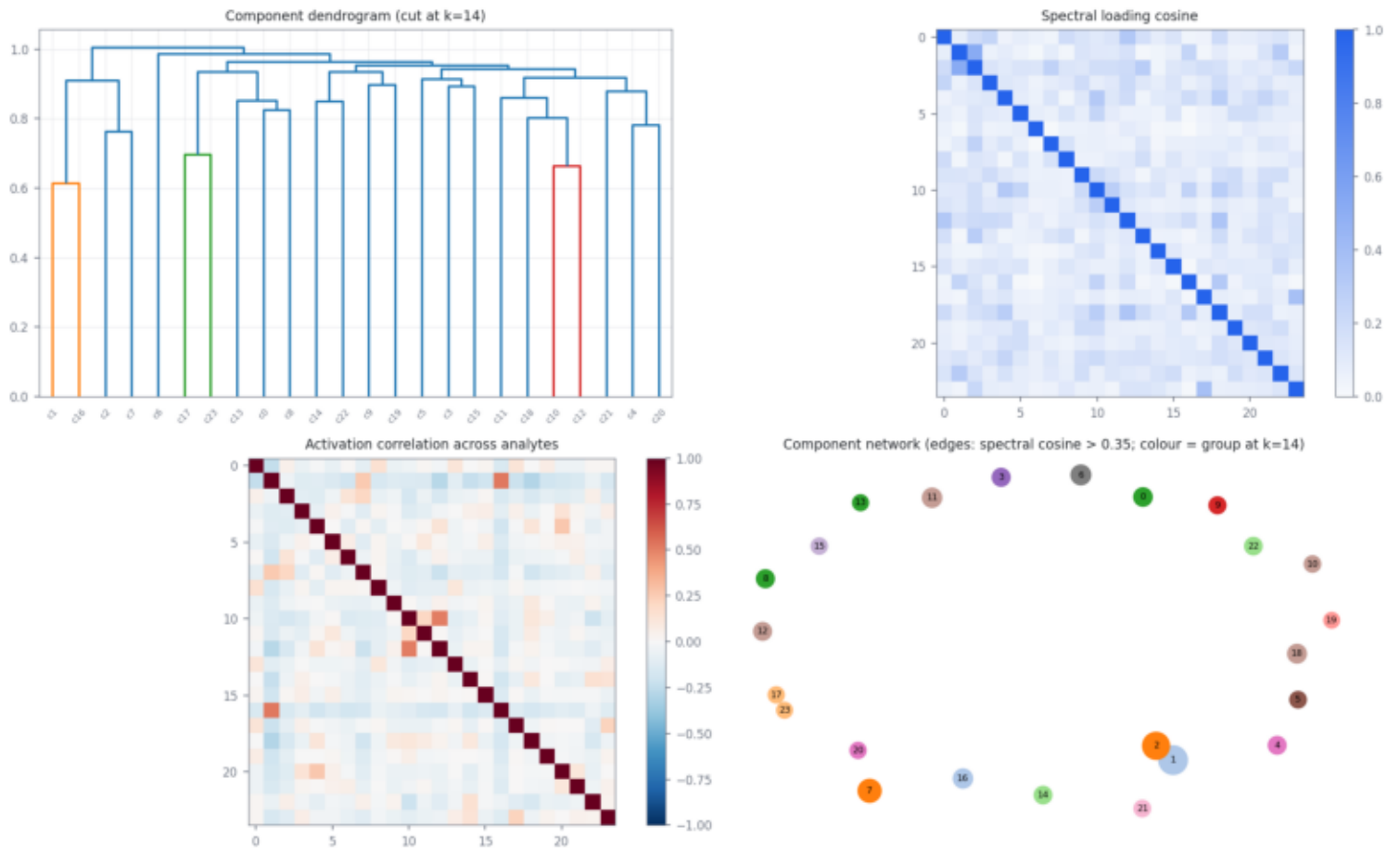
Keep all 24 components as the canonical frozen numerical layer; report biochemistry through a versioned theme overlay (Option C, hierarchical BSV); publish MSS at class level with explicit confusable-group annotations. Do not expand k, do not retrain, do not admit Ag-SERS.

Atlas overview



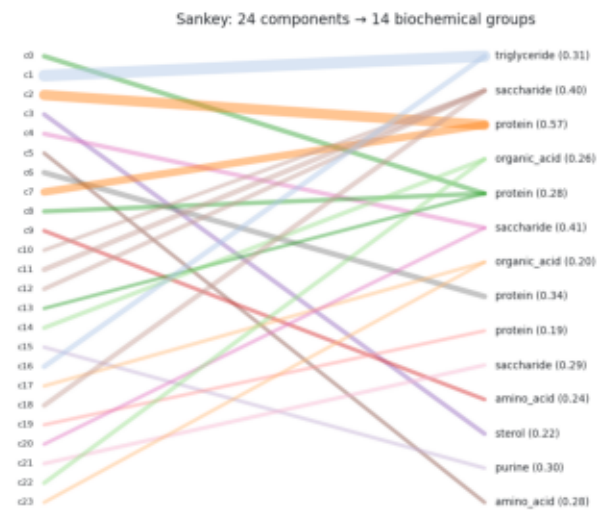
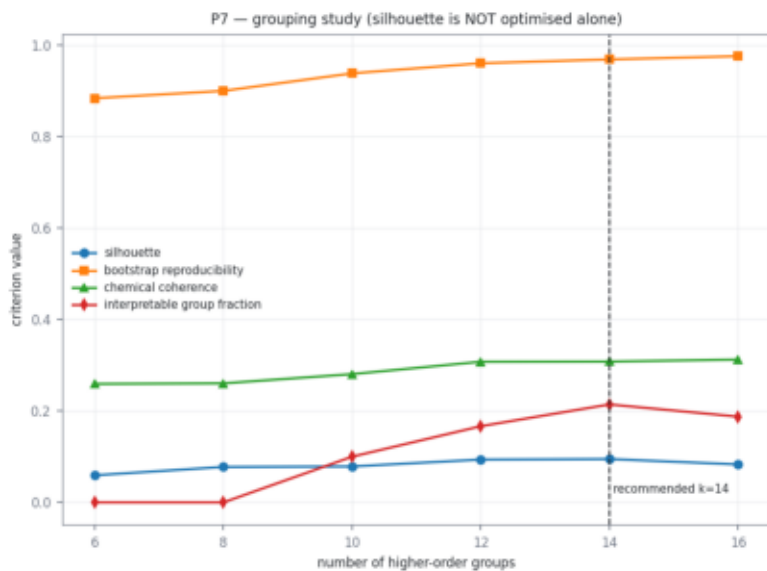
Atlas overview — component x chemical-family composition and the component x analyte matrix. Most components draw on several families: enrichment, not purity.

P6 component relationships



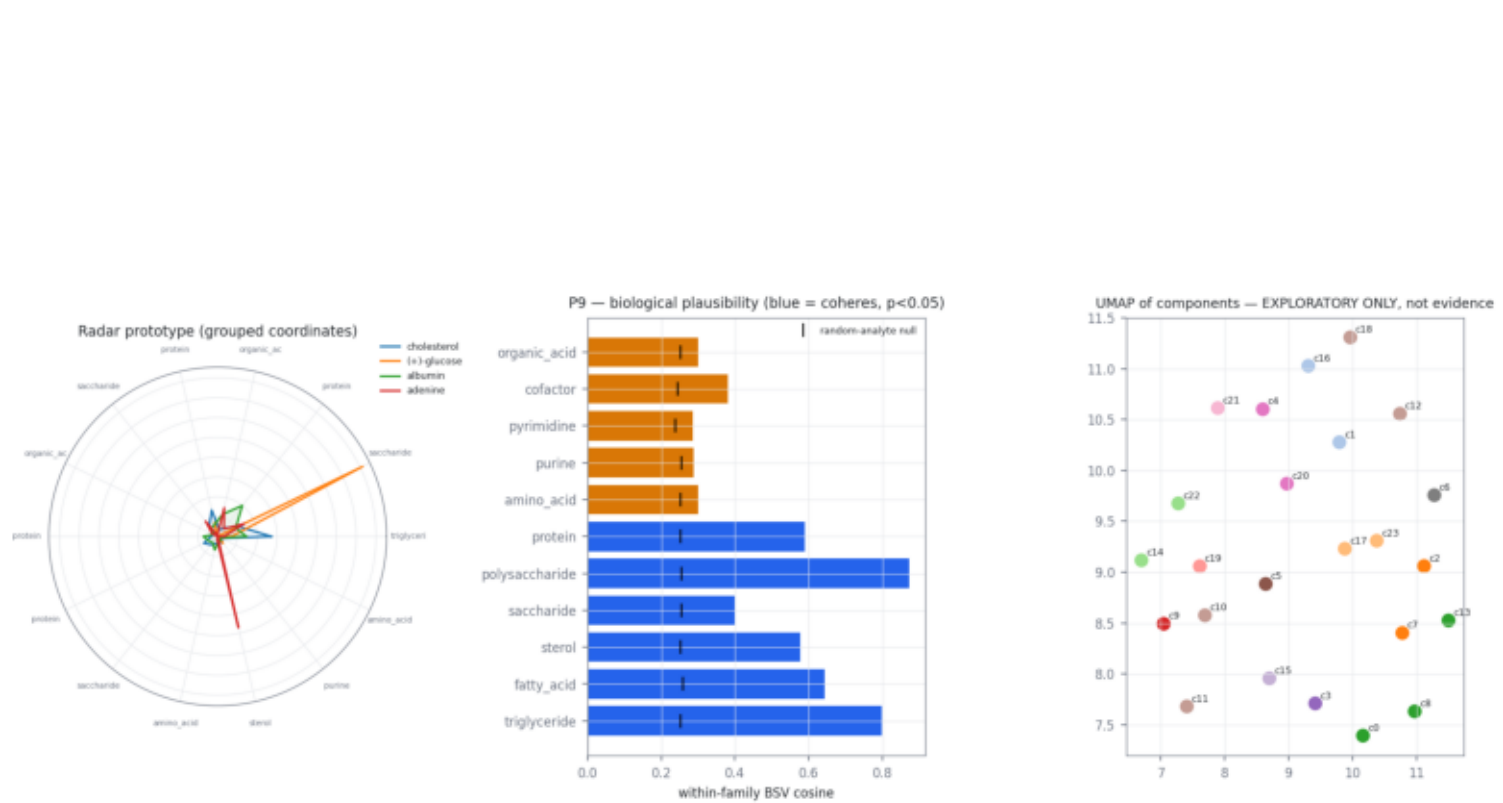
P6 component relationships — dendrogram, spectral-loading cosine, activation correlation and the component network. Off-diagonal structure is modest, consistent with NMF parts that are largely distinct.

P7 grouping study and the flow from 24 components to higher-order groups.



P7 grouping study and the flow from 24 components to higher-order groups. Silhouette stays low at every k, so grouping is an overlay, not a discovered partition.

Radar prototype on grouped coordinates; P9 biological plausibility against a r



Radar prototype on grouped coordinates; P9 biological plausibility against a random-analyte null;
UMAP of components marked exploratory only.

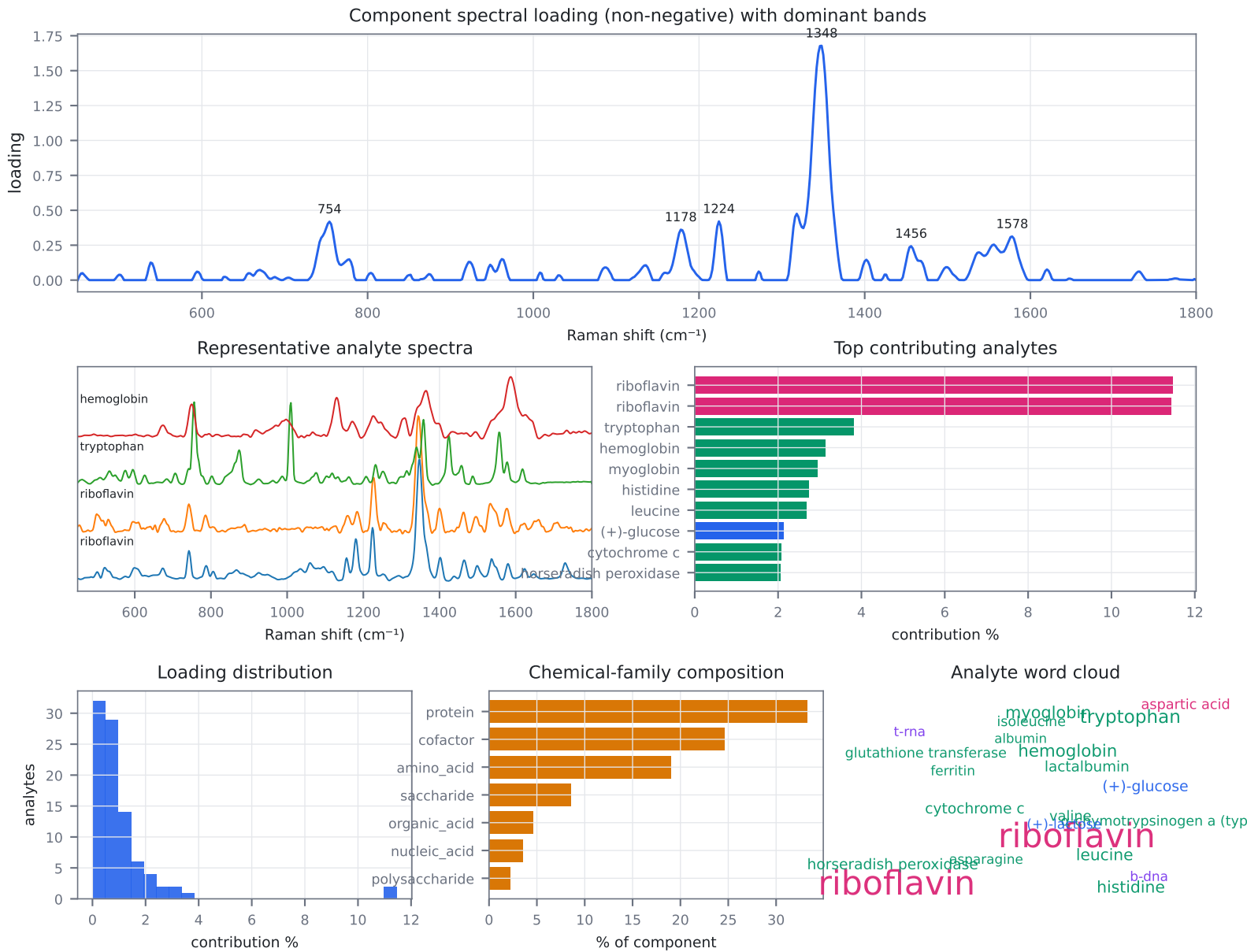
Atlas entries

one page per latent component (24)

Component 0

tentative theme: protein · class: protein/peptide · confidence: moderate

variance 0.0351 stability 0.810 purity 0.333 enrichment 1.74x entropy 0.849
analytes contributing 92 dominant (≥5%) 2 sparsity 0.821 band uniqueness 0.91 median band width 21 cm⁻¹
representative analyte: riboflavin · secondary family: amino_acid



Interpretation. The component is dominated by protein/peptide chemistry (family 'protein', 33% of its loading, 1.7x enriched over the corpus base rate). Its strongest bands lie at 754, 1178, 1224, 1348, 1456 cm⁻¹. Literature corroboration: {'NucleicAcids': 3, 'AminoAcids': 2, 'Proteins': 2}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

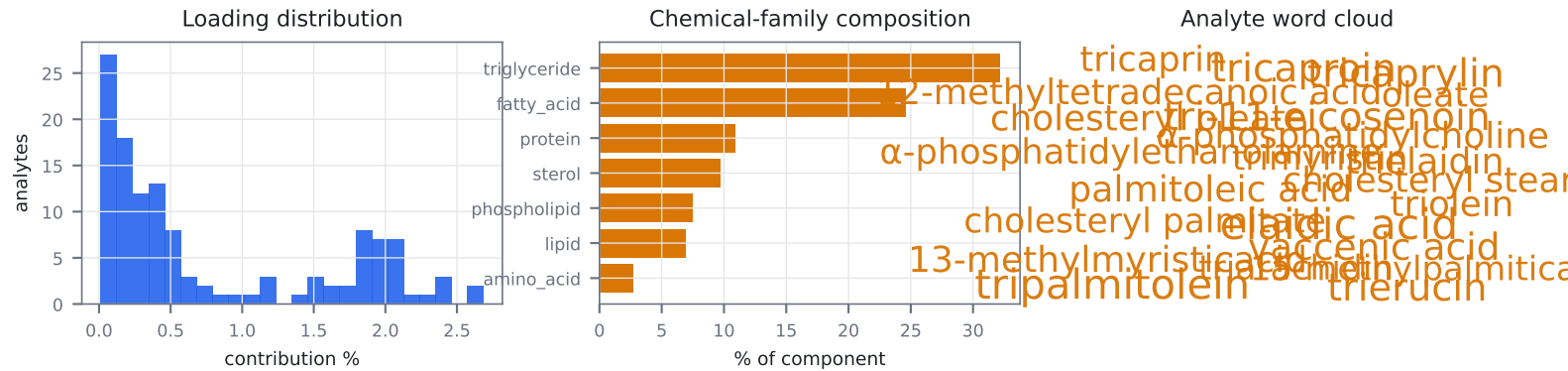
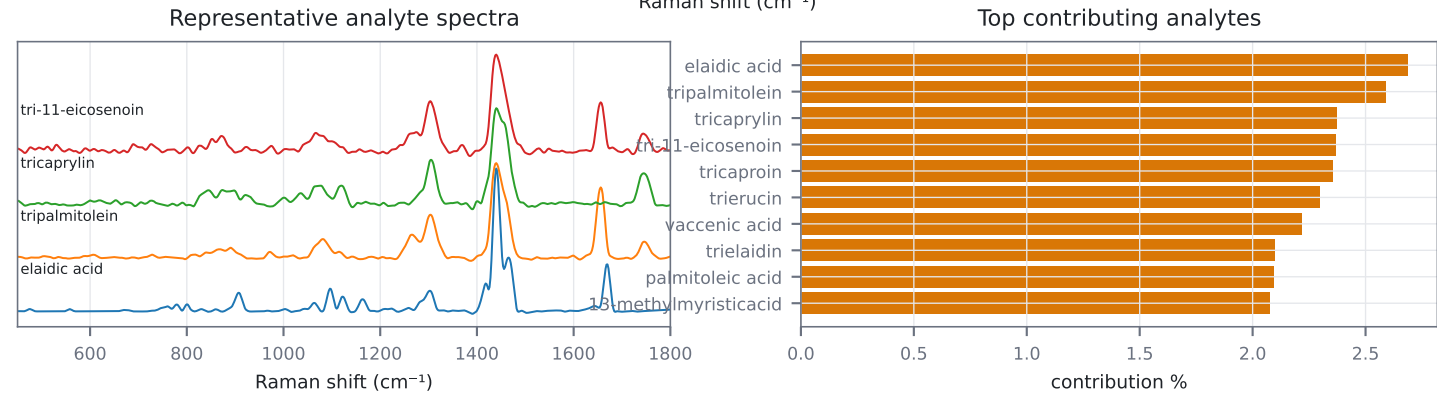
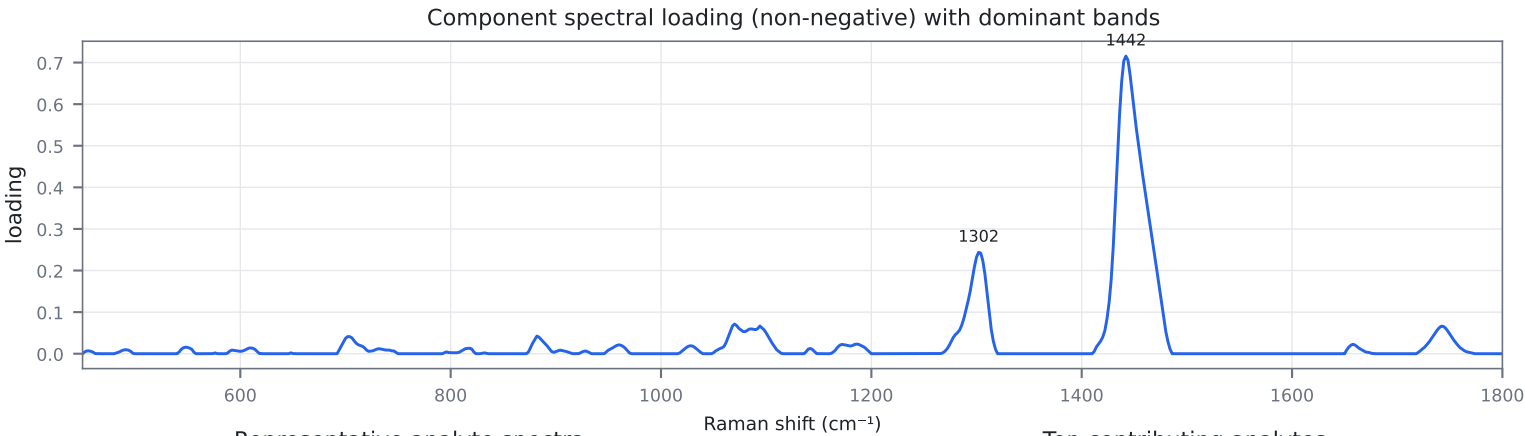
Full analyte composition — top 18 of 92 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
riboflavin	11.46	cofactor/small-molecule	cofactor	unavailable
riboflavin	11.43	cofactor/small-molecule	cofactor	unavailable
tryptophan	3.81	protein/peptide	amino_acid	aromatic amino acid
hemoglobin	3.12	protein/peptide	protein	unavailable
myoglobin	2.93	protein/peptide	protein	unavailable
histidine	2.73	protein/peptide	amino_acid	aromatic amino acid
leucine	2.67	protein/peptide	amino_acid	aliphatic/polar amino acid
(+)-glucose	2.12	carbohydrate	saccharide	monosaccharide
cytochrome c	2.07	protein/peptide	protein	unavailable
horseradish peroxidase	2.06	protein/peptide	protein	unavailable

Component 1

tentative theme: triglyceride · class: lipid · confidence: high

variance 0.1186 stability 0.955 purity 0.322 enrichment 3.36x entropy 0.888
analytes contributing 126 dominant (≥5%) 0 sparsity 0.888 band uniqueness 0.91 median band width 25 cm⁻¹
representative analyte: elaidic acid secondary family: triglyceride



Interpretation. The component is dominated by lipid chemistry (family 'triglyceride', 32% of its loading, 3.4x enriched over the corpus base rate). Its strongest bands lie at 1302, 1442 cm⁻¹. Literature corroboration: {'Lipids-FattyAcids': 2}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

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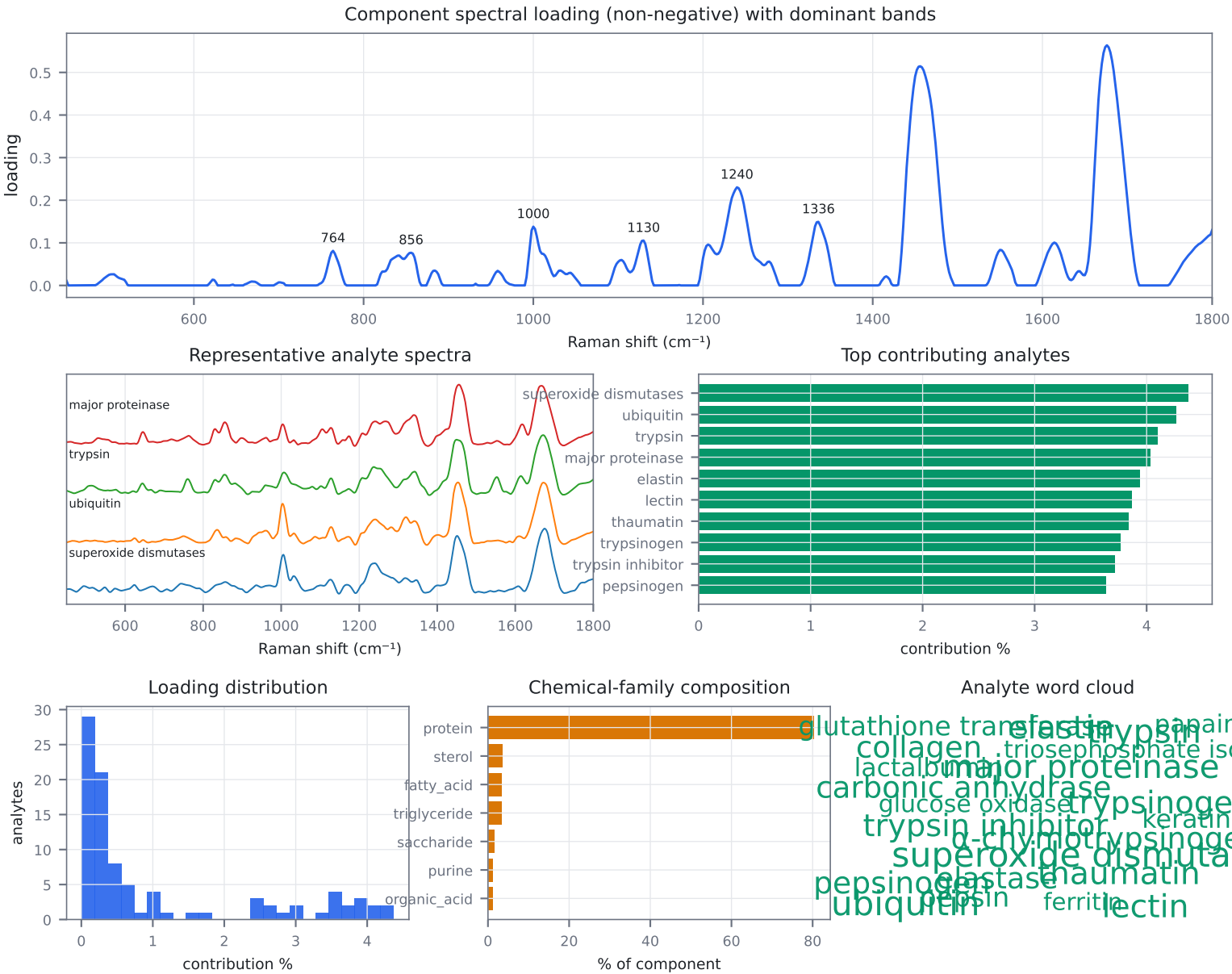
Full analyte composition — top 18 of 126 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
elaidic acid	2.68	lipid	fatty_acid	saturated fatty acid
tripalmitolein	2.59	lipid	triglyceride	triacylglycerol
tricaprylin	2.37	lipid	triglyceride	triacylglycerol
tri-11-eicosenoin	2.37	lipid	triglyceride	triacylglycerol
tricaproin	2.35	lipid	triglyceride	triacylglycerol
trierucin	2.29	lipid	triglyceride	triacylglycerol
vaccenic acid	2.22	lipid	fatty_acid	saturated fatty acid
trielaidin	2.1	lipid	triglyceride	triacylglycerol
palmitoleic acid	2.09	lipid	fatty_acid	unsaturated fatty acid
13-methylmyristic acid	2.08	lipid	fatty_acid	saturated fatty acid

Component 2

tentative theme: protein · class: protein/peptide · confidence: high

variance 0.1028 stability 0.965 purity 0.803 enrichment 4.19x entropy 0.836
analytes contributing 93 dominant (≥5%) 0 sparsity 0.773 band uniqueness 0.88 median band width 22 cm⁻¹
representative analyte: superoxide dismutases · secondary family: triglyceride



Interpretation. The component is dominated by protein/peptide chemistry (family 'protein', 80% of its loading, 4.2x enriched over the corpus base rate). Its strongest bands lie at 764, 856, 1000, 1130, 1240 cm⁻¹. Literature corroboration: {'Proteins': 6, 'NucleicAcids': 3, 'Lipids-FattyAcids': 1, 'AminoAcids': 1}.

Artifact assessment. Bootstrap stability is high and the component is chemically focused, so it is unlikely to be a fitting artifact.

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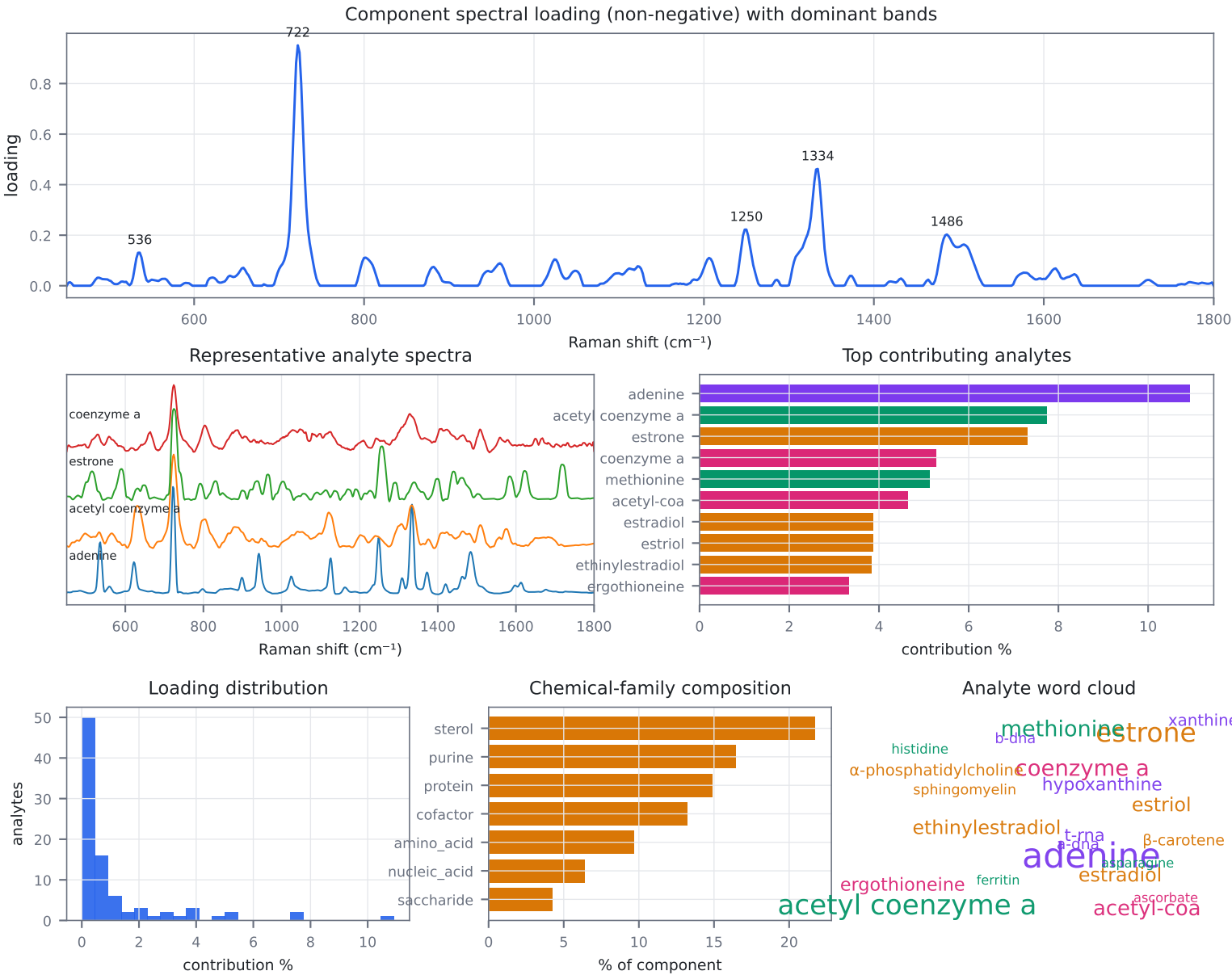
Full analyte composition — top 18 of 93 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
superoxide dismutases	4.37	protein/peptide	protein	unavailable
ubiquitin	4.26	protein/peptide	protein	unavailable
trypsin	4.1	protein/peptide	protein	unavailable
major proteinase	4.03	protein/peptide	protein	unavailable
elastin	3.94	protein/peptide	protein	unavailable
lectin	3.86	protein/peptide	protein	unavailable
thaumatin	3.84	protein/peptide	protein	unavailable
trypsinogen	3.76	protein/peptide	protein	unavailable
trypsin inhibitor	3.72	protein/peptide	protein	unavailable
pepsinogen	3.63	protein/peptide	protein	unavailable

Component 3

tentative theme: sterol · class: lipid · confidence: moderate

variance 0.0315 stability 0.823 purity 0.217 enrichment 4.03x entropy 0.800
analytes contributing 90 dominant (≥5%) 5 sparsity 0.759 band uniqueness 0.89 median band width 14 cm⁻¹
representative analyte: adenine · secondary family: saccharide



Interpretation. The component is dominated by lipid chemistry (family 'sterol', 22% of its loading, 4.0x enriched over the corpus base rate). Its strongest bands lie at 536, 722, 1250, 1334, 1486 cm⁻¹. Literature corroboration: {'NucleicAcids': 3, 'Proteins': 2, 'Lipids-FattyAcids': 1, 'AminoAcids': 1}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

Full analyte composition — top 18 of 90 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
adenine	10.92	nucleic	purine	purine base/derivative
acetyl coenzyme a	7.74	protein/peptide	protein	unavailable
estrone	7.31	lipid	sterol	unavailable
coenzyme a	5.26	cofactor/small-molecule	cofactor	unavailable
methionine	5.12	protein/peptide	amino_acid	aliphatic/polar amino acid
acetyl-coa	4.63	cofactor/small-molecule	cofactor	unavailable
estradiol	3.87	lipid	sterol	unavailable
estriol	3.86	lipid	sterol	unavailable
ethinylestradiol	3.84	lipid	sterol	unavailable
ergothioneine	3.32	cofactor/small-molecule	cofactor	unavailable

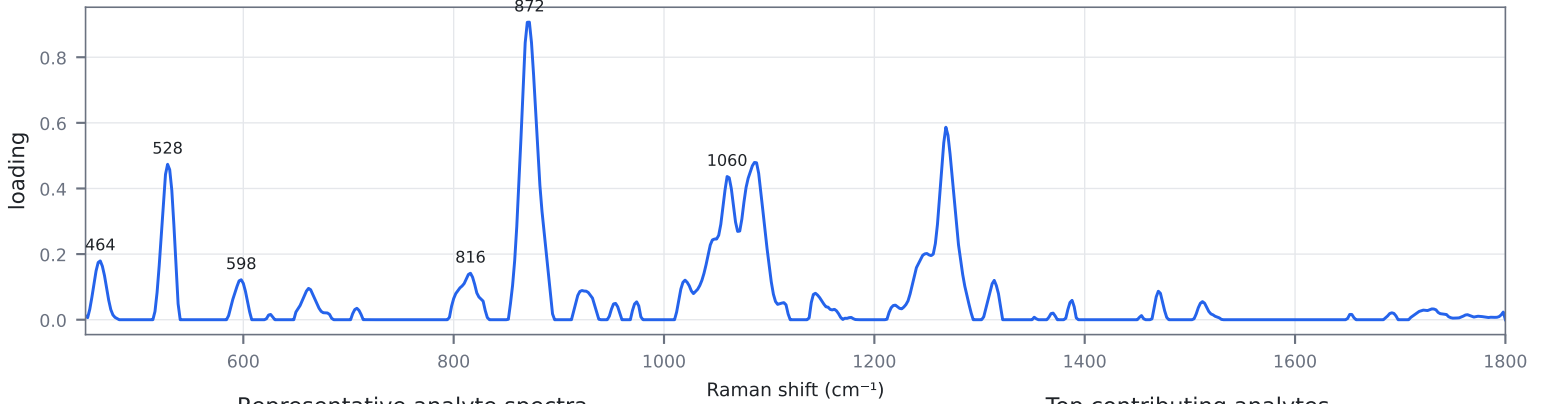
Component 4

tentative theme: saccharide · class: carbohydrate · confidence: high

variance 0.0309 stability 0.769 purity 0.429 enrichment 2.65x entropy 0.889

analytes contributing 115 dominant ($\geq 5\%$) 2 sparsity 0.807 band uniqueness 0.89 median band width 14 cm^{-1}

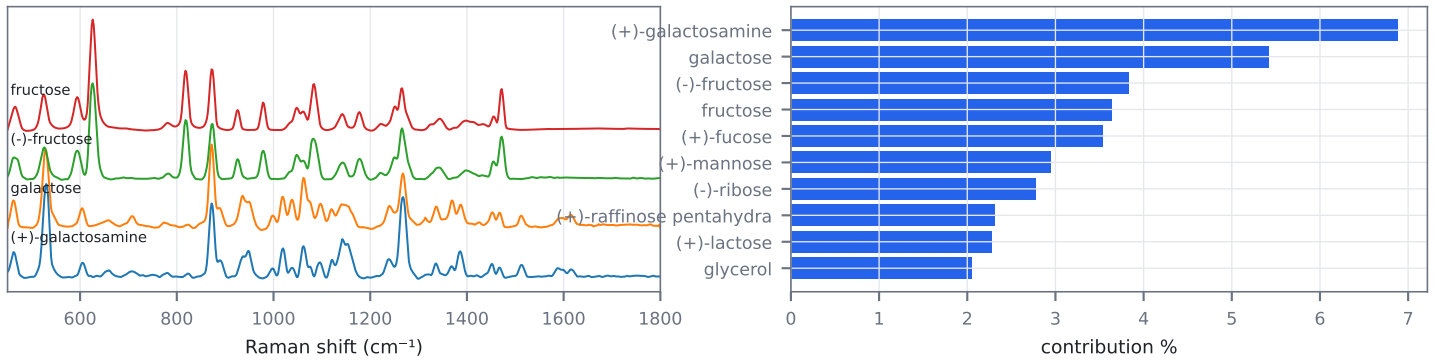
representative analyte: (+)-galactosamine · secondary family: saccharide



Representative analyte spectra

Raman shift (cm⁻¹)

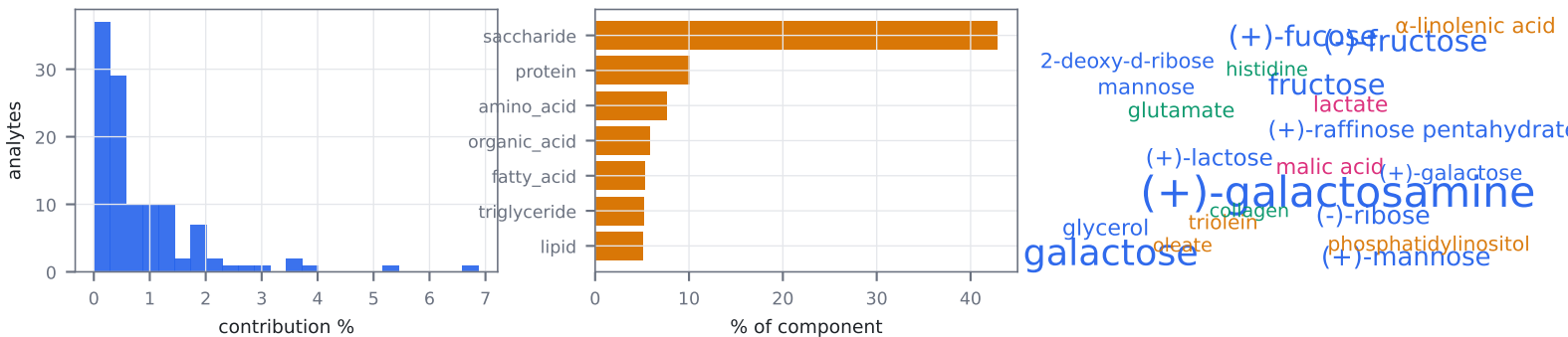
Top contributing analytes



Loading distribution

Chemical-family composition

Analyte word cloud



Interpretation. The component is dominated by carbohydrate chemistry (family 'saccharide', 43% of its loading, 2.7x enriched over the corpus base rate). Its strongest bands lie at 464, 528, 598, 816, 872 cm^{-1} .

Literature corroboration: {'Lipids-FattyAcids': 3, 'NucleicAcids': 2, 'Lipids-Hormones': 2, 'Saccharides-Monosaccharides': 1, 'Proteins': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

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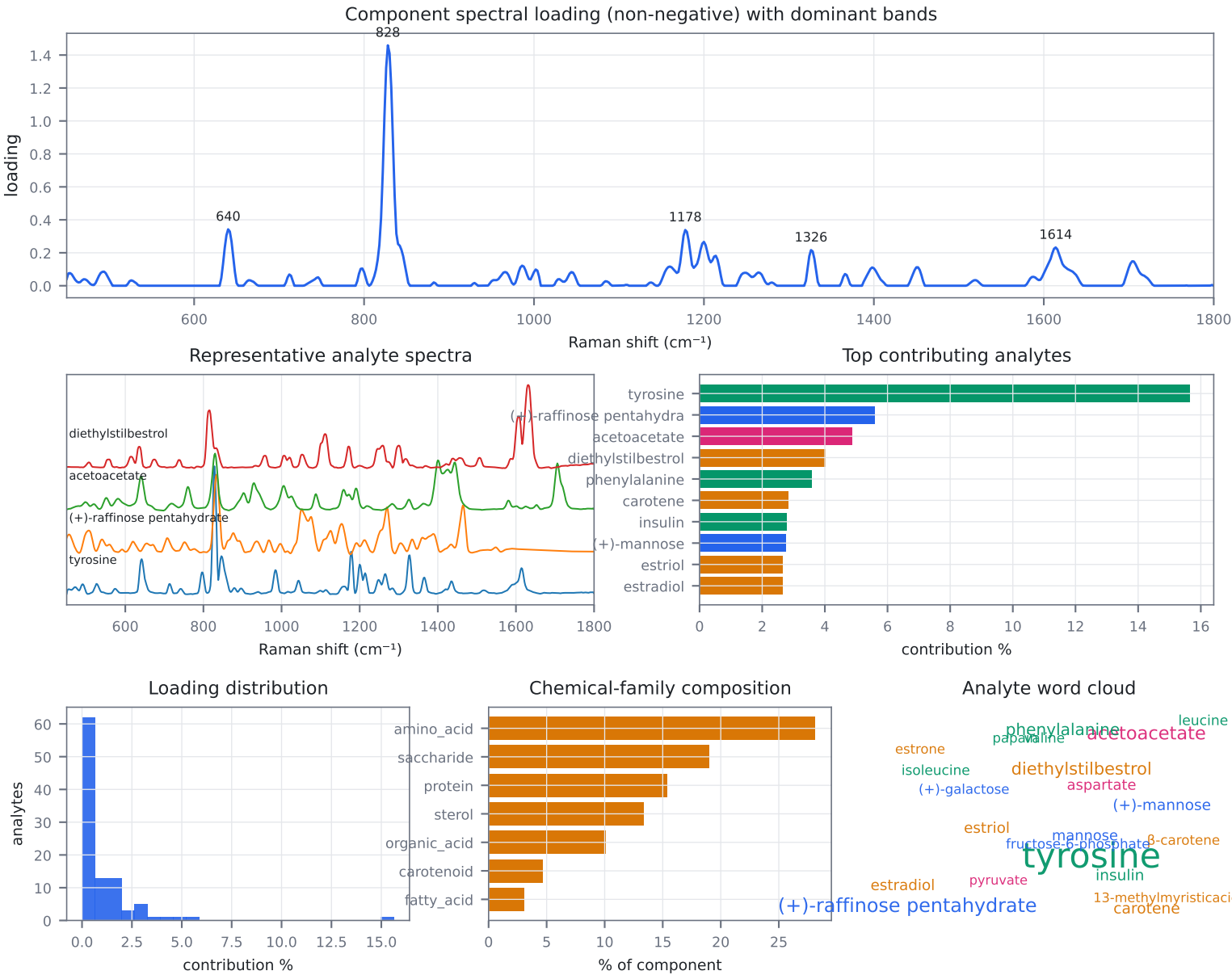
Full analyte composition — top 18 of 115 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
(+)-galactosamine	6.87	carbohydrate	saccharide	monosaccharide
galactose	5.42	carbohydrate	saccharide	disaccharide
(-)-fructose	3.83	carbohydrate	saccharide	monosaccharide
fructose	3.63	carbohydrate	saccharide	monosaccharide
(+)-fucose	3.54	carbohydrate	saccharide	monosaccharide
(+)-mannose	2.95	carbohydrate	saccharide	monosaccharide
(-)-ribose	2.77	carbohydrate	saccharide	monosaccharide
(+)-raffinose pentahydrate	2.31	carbohydrate	saccharide	trisaccharide
(+)-lactose	2.28	carbohydrate	saccharide	disaccharide
glycerol	2.05	carbohydrate	polyol	unavailable

Component 5

tentative theme: amino_acid · class: protein/peptide · confidence: moderate

variance 0.0237 stability 0.737 purity 0.281 enrichment 2.76x entropy 0.819
analytes contributing 101 dominant (≥5%) 2 sparsity 0.803 band uniqueness 0.87 median band width 12 cm⁻¹
representative analyte: tyrosine · secondary family: amino_acid



Interpretation. The component is dominated by protein/peptide chemistry (family 'amino_acid', 28% of its loading, 2.8x enriched over the corpus base rate). Its strongest bands lie at 640, 828, 1178, 1326, 1614 cm⁻¹. Literature corroboration: {'Proteins': 5, 'Lipids-FattyAcids': 2}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

Full analyte composition — top 18 of 101 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
tyrosine	15.63	protein/peptide	amino_acid	aromatic amino acid
(+)-raffinose pentahydrate	5.58	carbohydrate	saccharide	trisaccharide
acetoacetate	4.87	cofactor/small-molecule	organic_acid	unavailable
diethylstilbestrol	4.02	lipid	sterol	unavailable
phenylalanine	3.59	protein/peptide	amino_acid	aromatic amino acid
carotene	2.84	lipid	carotenoid	unavailable
insulin	2.77	protein/peptide	protein	unavailable
(+)-mannose	2.75	carbohydrate	saccharide	monosaccharide
estriol	2.66	lipid	sterol	unavailable
estradiol	2.65	lipid	sterol	unavailable

Component 6

tentative theme: protein · class: protein/peptide · confidence: moderate

variance 0.0439 stability 0.872 purity 0.335 enrichment 1.75x entropy 0.809
analytes contributing 109 dominant (≥5%) 3 sparsity 0.859 band uniqueness 0.96 median band width 16 cm⁻¹
representative analyte: urea · secondary family: saccharide



Interpretation. The component is dominated by protein/peptide chemistry (family 'protein', 33% of its loading, 1.7x enriched over the corpus base rate). Its strongest bands lie at 1006 cm⁻¹. Literature corroboration: {'AminoAcids': 1, 'NucleicAcids': 1}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

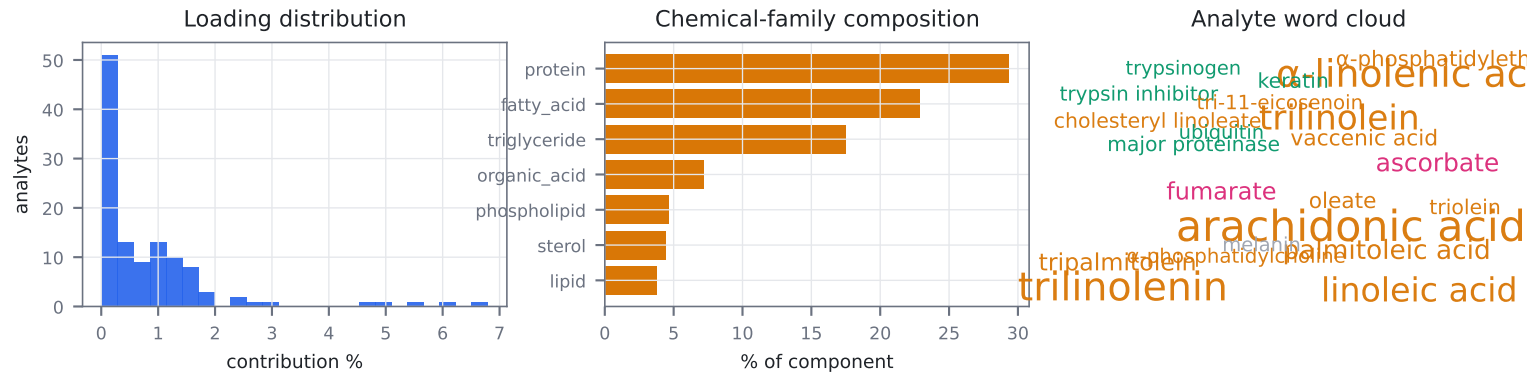
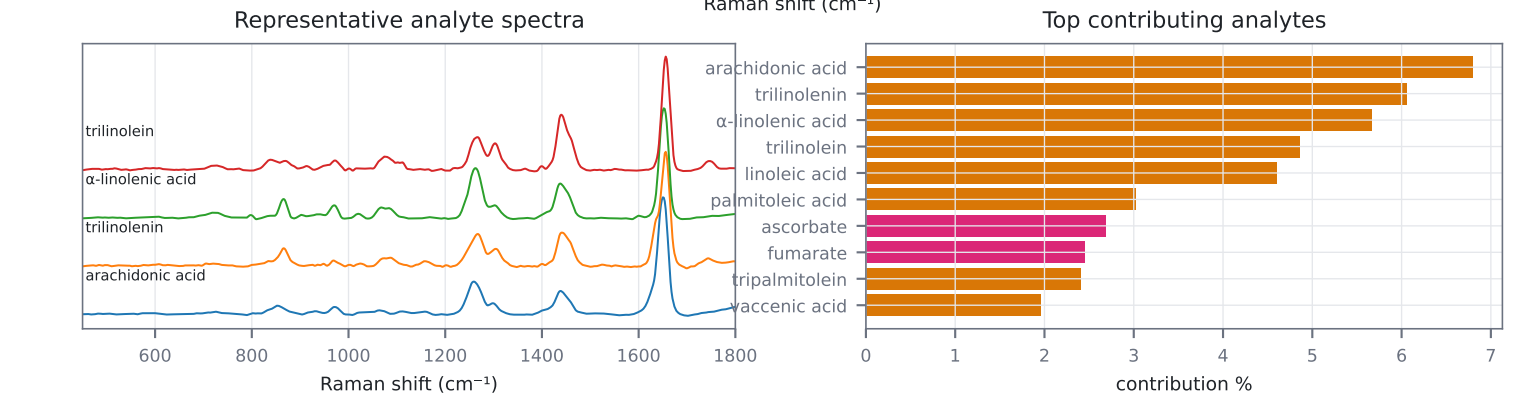
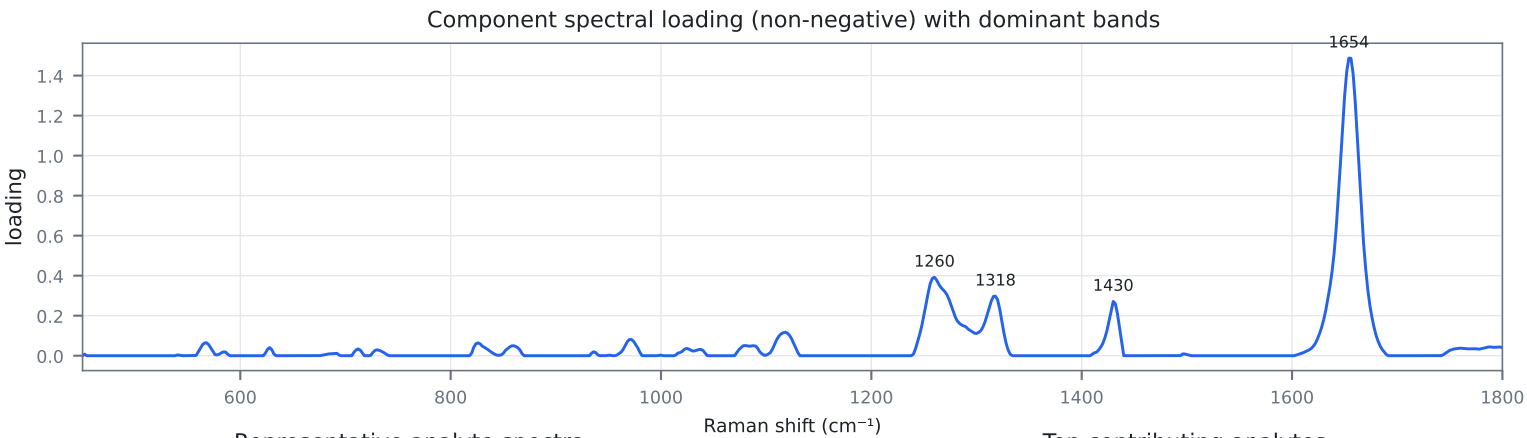
Full analyte composition — top 18 of 109 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
urea	11.87	cofactor/small-molecule	small_nitrogenous	unavailable
ure	11.17	unassigned	unknown	unavailable
phenylalanine	7.33	protein/peptide	amino_acid	aromatic amino acid
β-carotene	3.62	lipid	carotenoid	unavailable
carotene	3.39	lipid	carotenoid	unavailable
tryptophan	3.3	protein/peptide	amino_acid	aromatic amino acid
lectin	2.92	protein/peptide	protein	unavailable
carbonic anhydrase	2.65	protein/peptide	protein	unavailable
pepsinogen	2.16	protein/peptide	protein	unavailable
thaumatin	2.16	protein/peptide	protein	unavailable

Component 7

tentative theme: protein · class: lipid · confidence: moderate

variance 0.0679 stability 0.951 purity 0.293 enrichment 1.53x entropy 0.851
analytes contributing 116 dominant (≥5%) 3 sparsity 0.879 band uniqueness 0.85 median band width 18 cm⁻¹
representative analyte: arachidonic acid · secondary family: saccharide



Interpretation. The component is dominated by lipid chemistry (family 'protein', 29% of its loading, 1.5x enriched over the corpus base rate). Its strongest bands lie at 1260, 1318, 1430, 1654 cm⁻¹. Literature corroboration: { 'Lipids-FattyAcids': 1, 'Proteins': 1 }.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

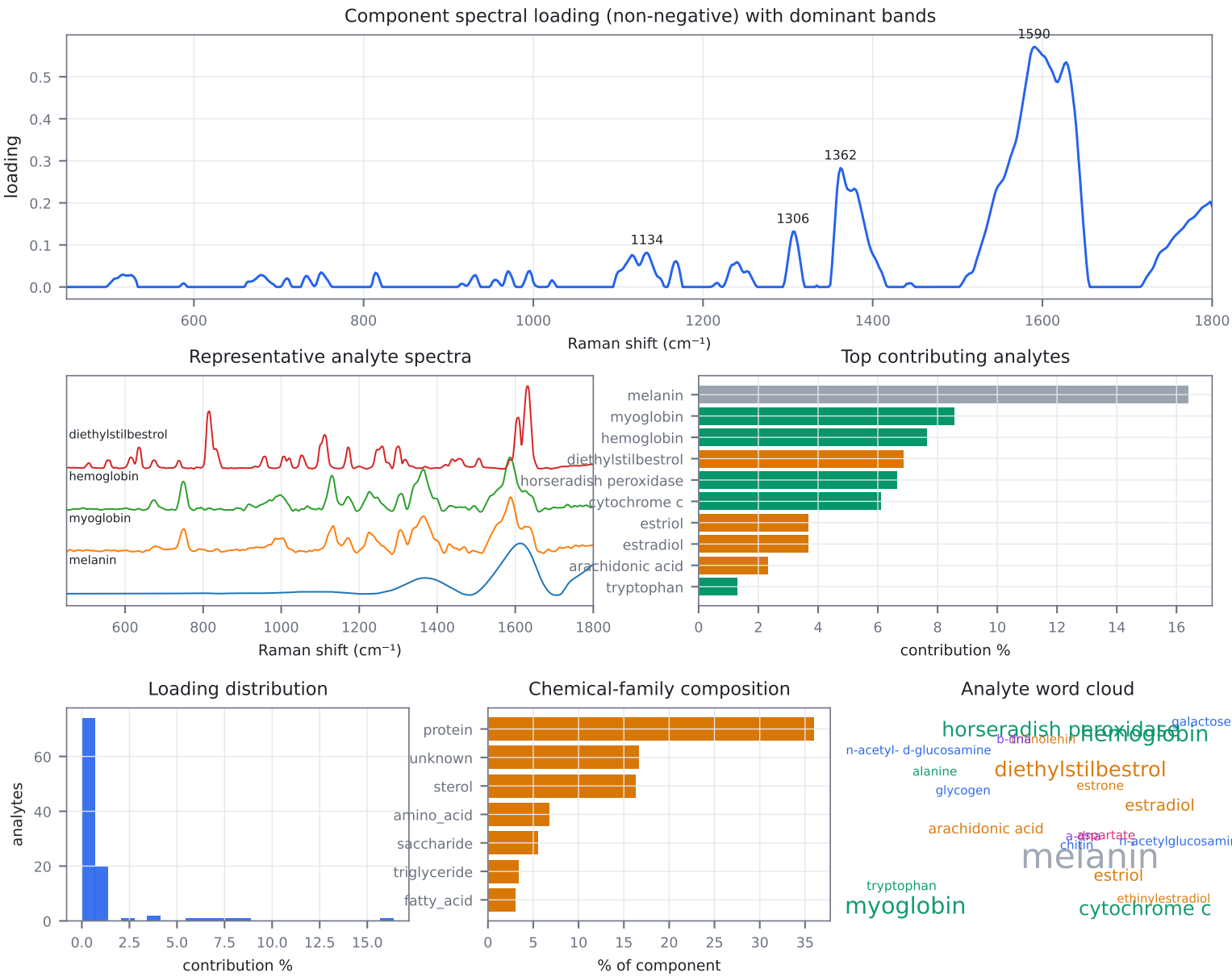
Full analyte composition — top 18 of 116 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
arachidonic acid	6.79	lipid	fatty_acid	unsaturated fatty acid
trilinolenin	6.05	lipid	triglyceride	triacylglycerol
α-linolenic acid	5.66	lipid	fatty_acid	unsaturated fatty acid
trilinolein	4.85	lipid	triglyceride	triacylglycerol
linoleic acid	4.6	lipid	fatty_acid	unsaturated fatty acid
palmitoleic acid	3.02	lipid	fatty_acid	unsaturated fatty acid
ascorbate	2.68	cofactor/small-molecule	organic_acid	unavailable
fumarate	2.45	cofactor/small-molecule	organic_acid	unavailable
tripalmitolein	2.4	lipid	triglyceride	triacylglycerol
vaccenic acid	1.96	lipid	fatty_acid	saturated fatty acid

Component 8

tentative theme: protein · class: protein/peptide · confidence: moderate

variance 0.0314 stability 0.927 purity 0.360 enrichment 1.88x entropy 0.763
analytes contributing 103 dominant (≥5%) 6 sparsity 0.812 band uniqueness 0.90 median band width 37 cm⁻¹
representative analyte: melanin · secondary family: amino_acid



Interpretation. The component is dominated by protein/peptide chemistry (family 'protein', 36% of its loading, 1.9x enriched over the corpus base rate). Its strongest bands lie at 1134, 1306, 1362, 1590 cm⁻¹. Literature corroboration: {'Lipids-FattyAcids': 2}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

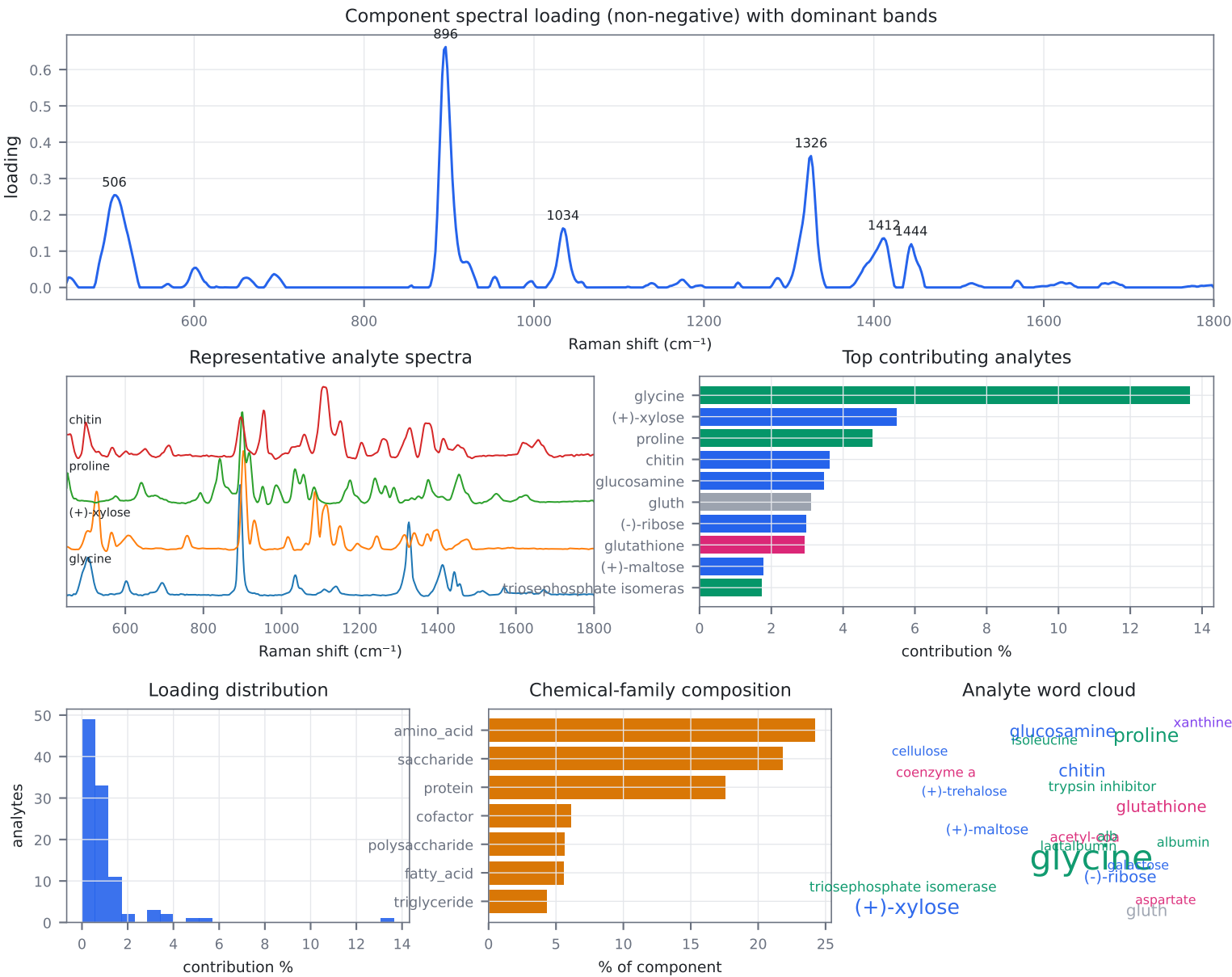
Full analyte composition — top 18 of 103 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
melanin	16.37	unassigned	unknown	unavailable
myoglobin	8.54	protein/peptide	protein	unavailable
hemoglobin	7.64	protein/peptide	protein	unavailable
diethylstilbestrol	6.85	lipid	sterol	unavailable
horseradish peroxidase	6.64	protein/peptide	protein	unavailable
cytochrome c	6.1	protein/peptide	protein	unavailable
estriol	3.67	lipid	sterol	unavailable
estradiol	3.67	lipid	sterol	unavailable
arachidonic acid	2.31	lipid	fatty_acid	unsaturated fatty acid
tryptophan	1.28	protein/peptide	amino_acid	aromatic amino acid

Component 9

tentative theme: amino_acid · class: protein/peptide · confidence: moderate

variance 0.0257 stability 0.709 purity 0.242 enrichment 2.38x entropy 0.868
analytes contributing 103 dominant (≥5%) 2 sparsity 0.849 band uniqueness 0.89 median band width 16 cm⁻¹
representative analyte: glycine · secondary family: saccharide



Interpretation. The component is dominated by protein/peptide chemistry (family 'amino_acid', 24% of its loading, 2.4x enriched over the corpus base rate). Its strongest bands lie at 506, 896, 1034, 1326, 1412 cm⁻¹. Literature corroboration: {'Proteins': 3, 'AminoAcids': 2, 'Saccharides-Monosaccharides': 2, 'NucleicAcids': 1, 'Lipids-FattyAcids': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

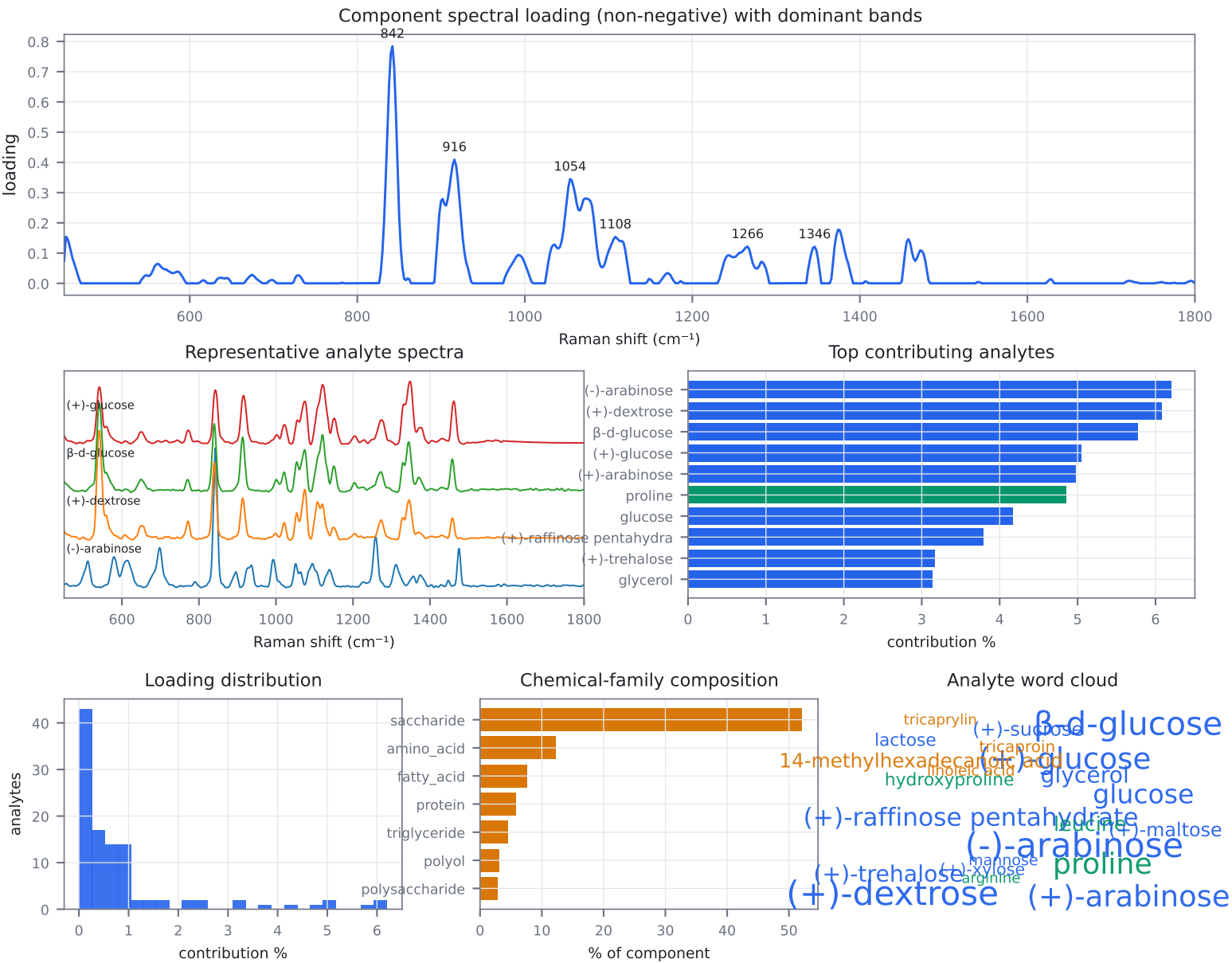
Full analyte composition — top 18 of 103 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
glycine	13.64	protein/peptide	amino_acid	aliphatic/polar amino acid
(+)-xylose	5.49	carbohydrate	saccharide	monosaccharide
proline	4.81	protein/peptide	amino_acid	aliphatic/polar amino acid
chitin	3.61	carbohydrate	polysaccharide	unavailable
glucosamine	3.45	carbohydrate	saccharide	monosaccharide
gluth	3.1	unassigned	unknown	unavailable
(-)-ribose	2.96	carbohydrate	saccharide	monosaccharide
glutathione	2.92	cofactor/small-molecule	cofactor	unavailable
(+)-maltose	1.76	carbohydrate	saccharide	disaccharide
triosephosphate isomerase	1.73	protein/peptide	protein	unavailable

Component 10

tentative theme: saccharide · class: carbohydrate · confidence: high

variance 0.0229 stability 0.653 purity 0.521 enrichment 3.22x entropy 0.843
analytes contributing 108 dominant (≥5%) 4 sparsity 0.817 band uniqueness 0.86 median band width 23 cm⁻¹
representative analyte: (-)-arabinose secondary family: saccharide



Interpretation. The component is dominated by carbohydrate chemistry (family 'saccharide', 52% of its loading, 3.2x enriched over the corpus base rate). Its strongest bands lie at 842, 916, 1054, 1108, 1266 cm⁻¹. Literature corroboration: {'Proteins': 2, 'NucleicAcids': 2, 'Saccharides-Polysaccharides': 1, 'Lipids-FattyAcids': 1, 'Lipids-Hormones': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

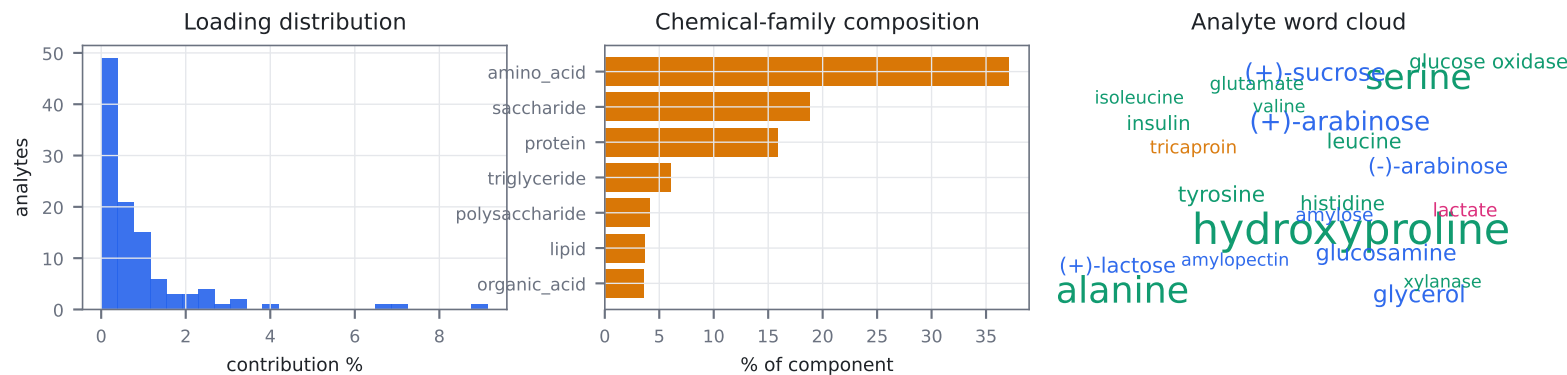
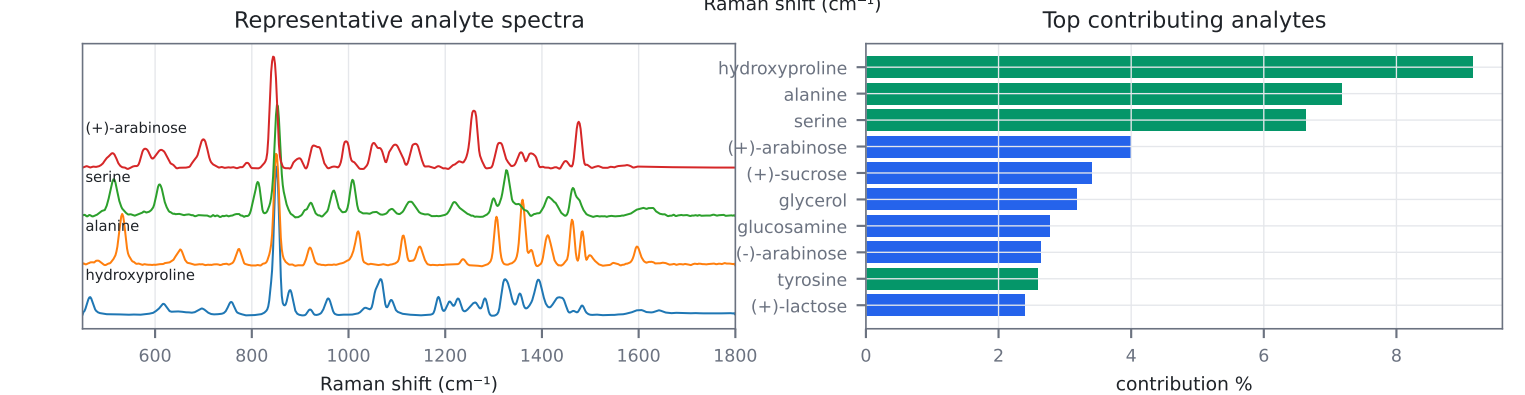
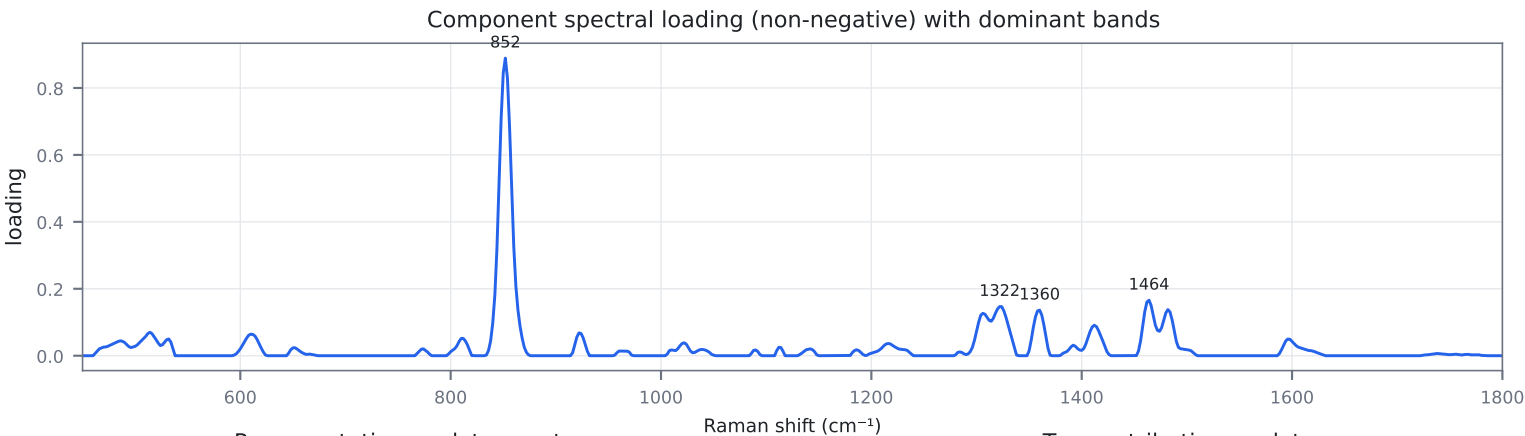
Full analyte composition — top 18 of 108 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
(-)-arabinose	6.2	carbohydrate	saccharide	monosaccharide
(+)-dextrose	6.08	carbohydrate	saccharide	monosaccharide
β-d-glucose	5.77	carbohydrate	saccharide	monosaccharide
(+)-glucose	5.05	carbohydrate	saccharide	monosaccharide
(+)-arabinose	4.97	carbohydrate	saccharide	monosaccharide
proline	4.85	protein/peptide	amino_acid	aliphatic/polar amino acid
glucose	4.17	carbohydrate	saccharide	monosaccharide
(+)-raffinose pentahydrate	3.79	carbohydrate	saccharide	trisaccharide
(+)-trehalose	3.16	carbohydrate	saccharide	disaccharide
glycerol	3.13	carbohydrate	polyol	unavailable

Component 11

tentative theme: amino_acid · class: protein/peptide · confidence: high

variance 0.0403 stability 0.886 purity 0.371 enrichment 3.64x entropy 0.851
analytes contributing 108 dominant (≥5%) 3 sparsity 0.828 band uniqueness 0.78 median band width 13 cm⁻¹
representative analyte: hydroxyproline · secondary family: saccharide



Interpretation. The component is dominated by protein/peptide chemistry (family 'amino_acid', 37% of its loading, 3.6x enriched over the corpus base rate). Its strongest bands lie at 852, 1322, 1360, 1464 cm⁻¹. Literature corroboration: {'Proteins': 4, 'Lipids-FattyAcids': 2}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

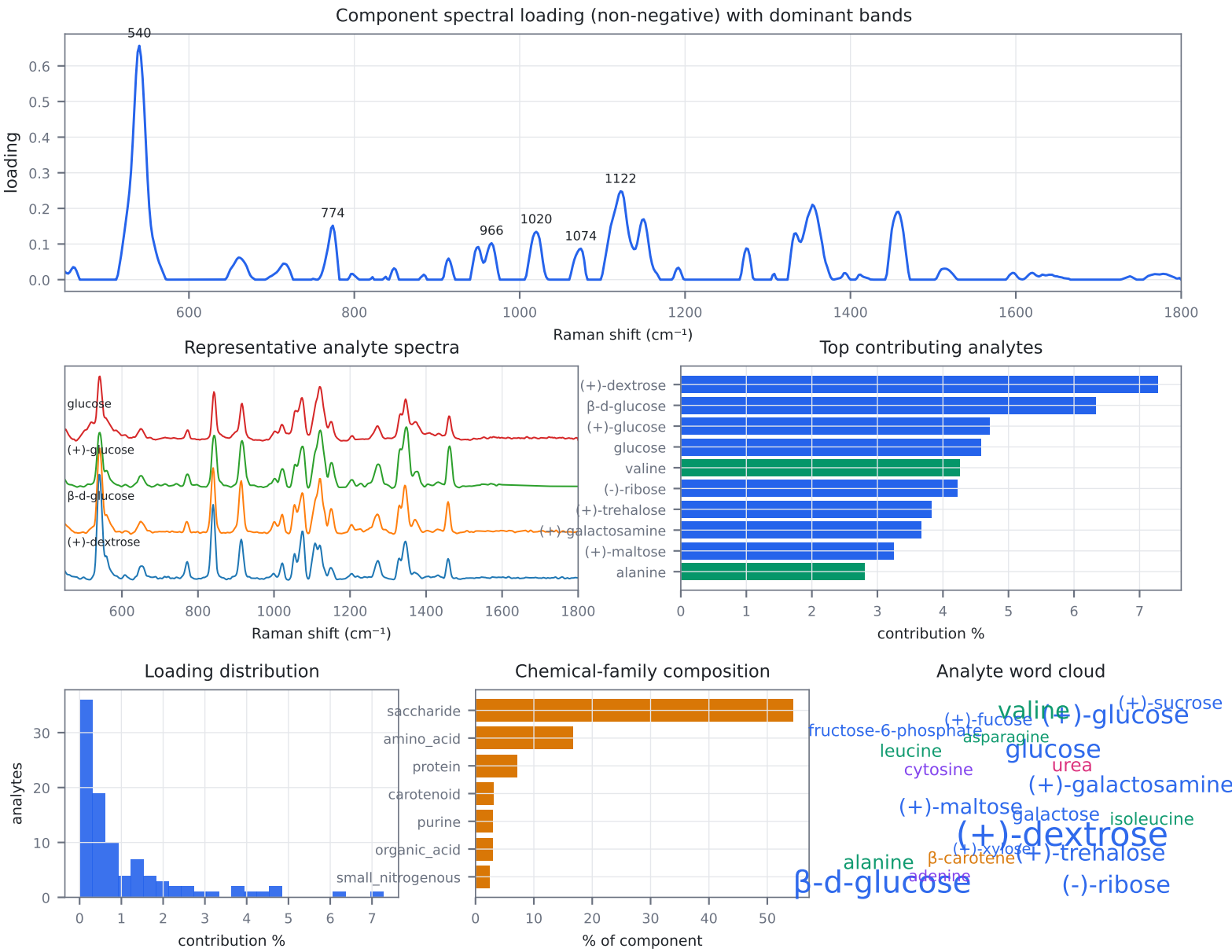
Full analyte composition — top 18 of 108 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
hydroxyproline	9.14	protein/peptide	amino_acid	aliphatic/polar amino acid
alanine	7.18	protein/peptide	amino_acid	aliphatic/polar amino acid
serine	6.64	protein/peptide	amino_acid	aliphatic/polar amino acid
(+)-arabinose	3.99	carbohydrate	saccharide	monosaccharide
(+)-sucrose	3.41	carbohydrate	saccharide	disaccharide
glycerol	3.17	carbohydrate	polyol	unavailable
glucosamine	2.77	carbohydrate	saccharide	monosaccharide
(-)-arabinose	2.64	carbohydrate	saccharide	monosaccharide
tyrosine	2.59	protein/peptide	amino_acid	aromatic amino acid
(+)-lactose	2.39	carbohydrate	saccharide	disaccharide

Component 12

tentative theme: saccharide · class: carbohydrate · confidence: high

variance 0.0313 stability 0.788 purity 0.544 enrichment 3.37x entropy 0.848
analytes contributing 97 dominant (≥5%) 2 sparsity 0.805 band uniqueness 0.90 median band width 15 cm⁻¹
representative analyte: (+)-dextrose secondary family: saccharide



Interpretation. The component is dominated by carbohydrate chemistry (family 'saccharide', 54% of its loading, 3.4x enriched over the corpus base rate). Its strongest bands lie at 540, 774, 966, 1020, 1074 cm⁻¹. Literature corroboration: {'Lipids-FattyAcids': 4, 'Proteins': 3, 'NucleicAcids': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

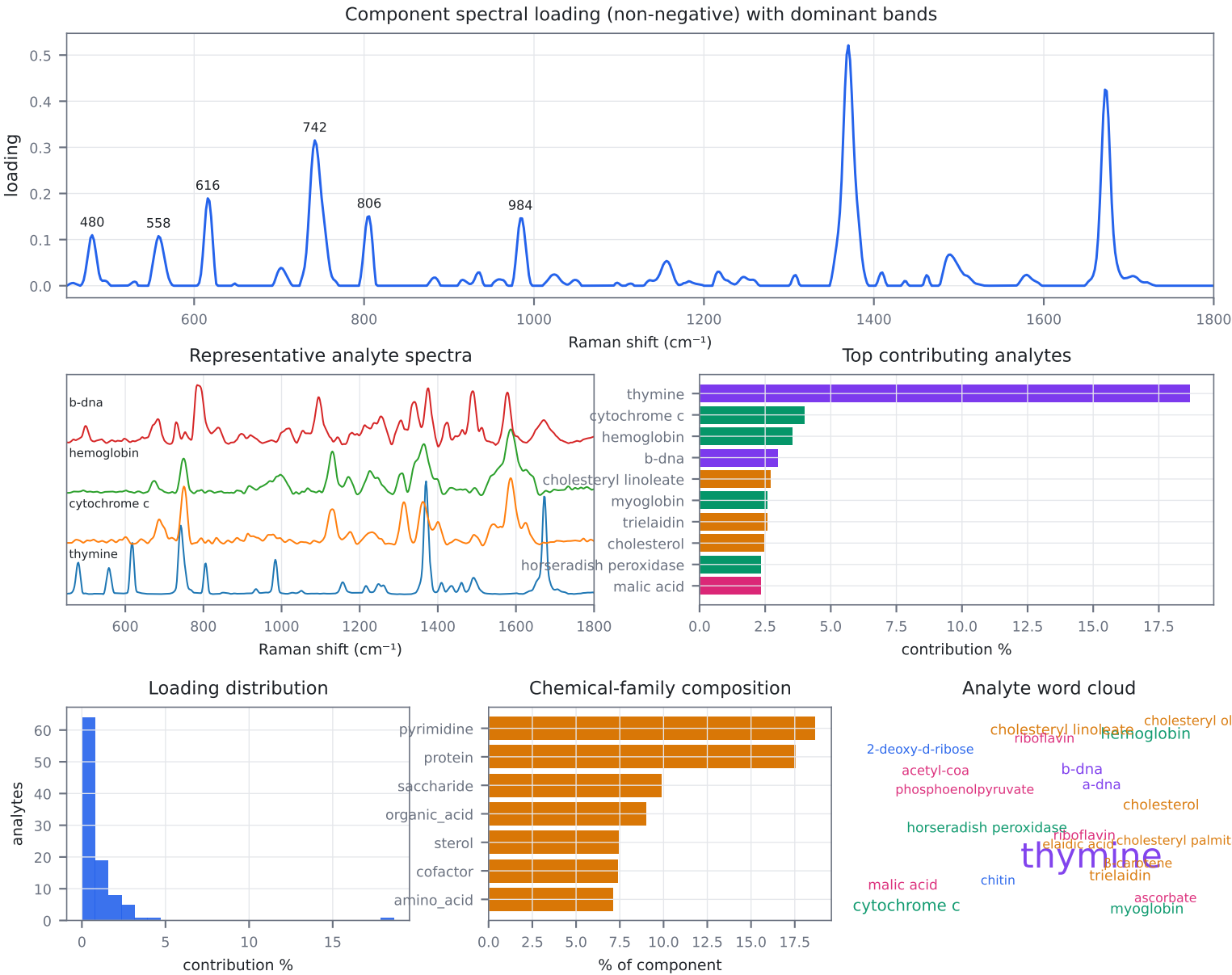
Full analyte composition — top 18 of 97 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
(+)-dextrose	7.27	carbohydrate	saccharide	monosaccharide
β-d-glucose	6.33	carbohydrate	saccharide	monosaccharide
(+)-glucose	4.71	carbohydrate	saccharide	monosaccharide
glucose	4.58	carbohydrate	saccharide	monosaccharide
valine	4.26	protein/peptide	amino_acid	aliphatic/polar amino acid
(-)-ribose	4.21	carbohydrate	saccharide	monosaccharide
(+)-trehalose	3.83	carbohydrate	saccharide	disaccharide
(+)-galactosamine	3.67	carbohydrate	saccharide	monosaccharide
(+)-maltose	3.25	carbohydrate	saccharide	disaccharide
alanine	2.81	protein/peptide	amino_acid	aliphatic/polar amino acid

Component 13

tentative theme: pyrimidine · class: nucleic · confidence: moderate

variance 0.0197 stability 0.664 purity 0.187 enrichment 10.39x entropy 0.834
analytes contributing 99 dominant (≥5%) 1 sparsity 0.831 band uniqueness 0.92 median band width 13 cm⁻¹
representative analyte: thymine · secondary family: saccharide



Interpretation. The component is dominated by nucleic chemistry (family 'pyrimidine', 19% of its loading, 10.4x enriched over the corpus base rate). Its strongest bands lie at 480, 558, 616, 742, 806 cm⁻¹.

Literature corroboration: {'Proteins': 8, 'AminoAcids': 2, 'Lipids-FattyAcids': 1, 'Saccharides-Monosaccharides': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

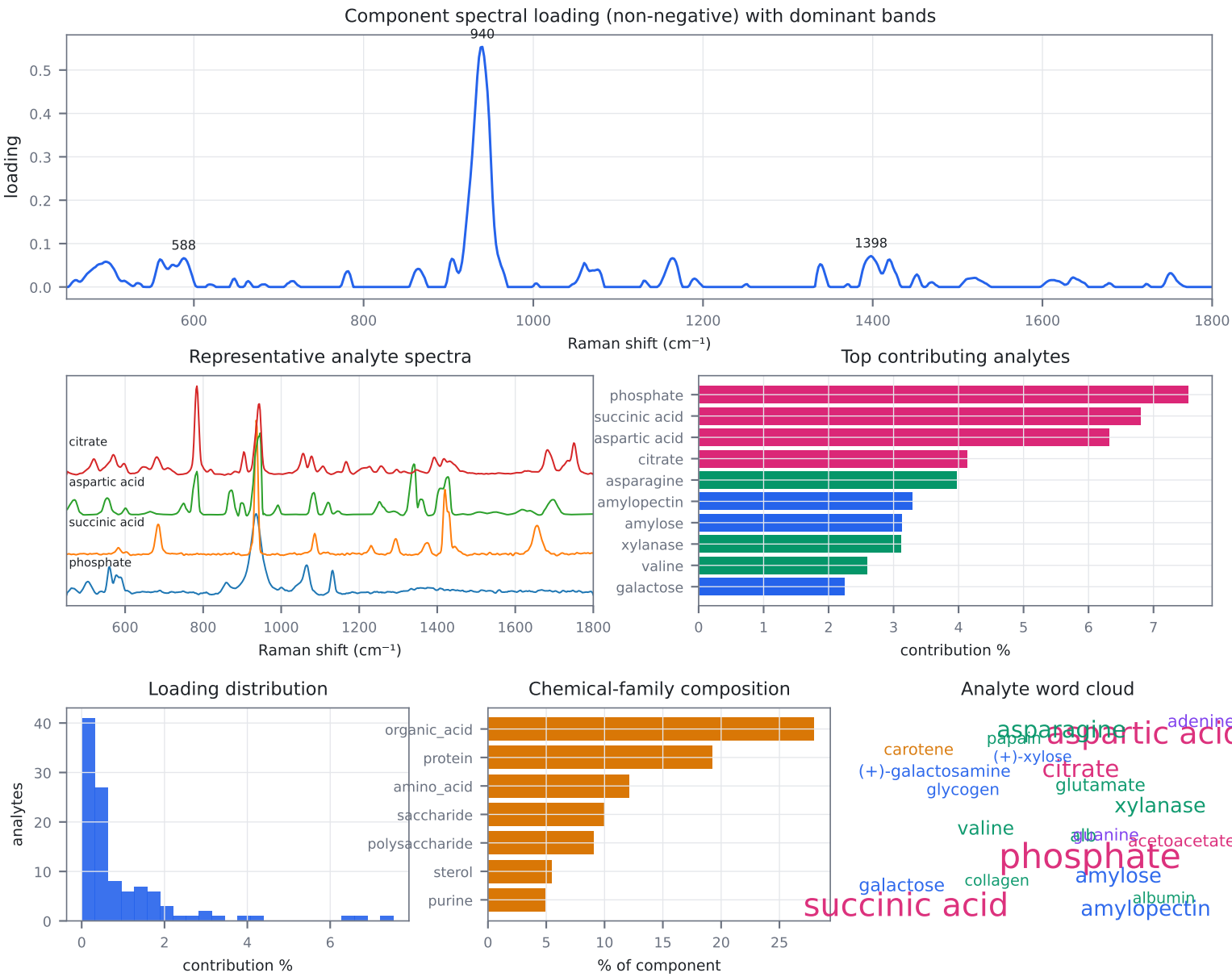
Full analyte composition — top 18 of 99 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
thymine	18.66	nucleic	pyrimidine	pyrimidine base
cytochrome c	4.0	protein/peptide	protein	unavailable
hemoglobin	3.54	protein/peptide	protein	unavailable
b-dna	2.96	nucleic	nucleic_acid	unavailable
cholesteryl linoleate	2.69	lipid	sterol	unavailable
myoglobin	2.59	protein/peptide	protein	unavailable
trielaidin	2.57	lipid	triglyceride	triacylglycerol
cholesterol	2.46	lipid	lipid	unavailable
horseradish peroxidase	2.34	protein/peptide	protein	unavailable
malic acid	2.34	cofactor/small molecule	organic_acid	unavailable

Component 14

tentative theme: organic_acid · class: protein/peptide · confidence: high

variance 0.0295 stability 0.769 purity 0.280 enrichment 3.89x entropy 0.849
analytes contributing 108 dominant (≥5%) 3 sparsity 0.826 band uniqueness 0.88 median band width 23 cm⁻¹
representative analyte: phosphate secondary family: saccharide



Interpretation. The component is dominated by protein/peptide chemistry (family 'organic_acid', 28% of its loading, 3.9x enriched over the corpus base rate). Its strongest bands lie at 588, 940, 1398 cm⁻¹. Literature corroboration: {'Saccharides-Monosaccharides': 2, 'Lipids-FattyAcids': 2, 'Saccharides-Polysaccharides': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

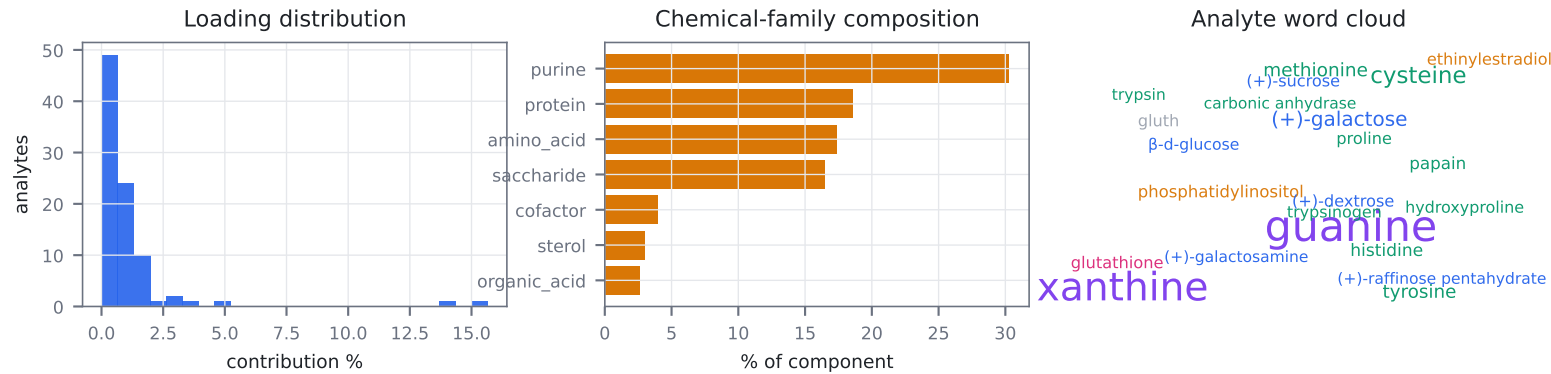
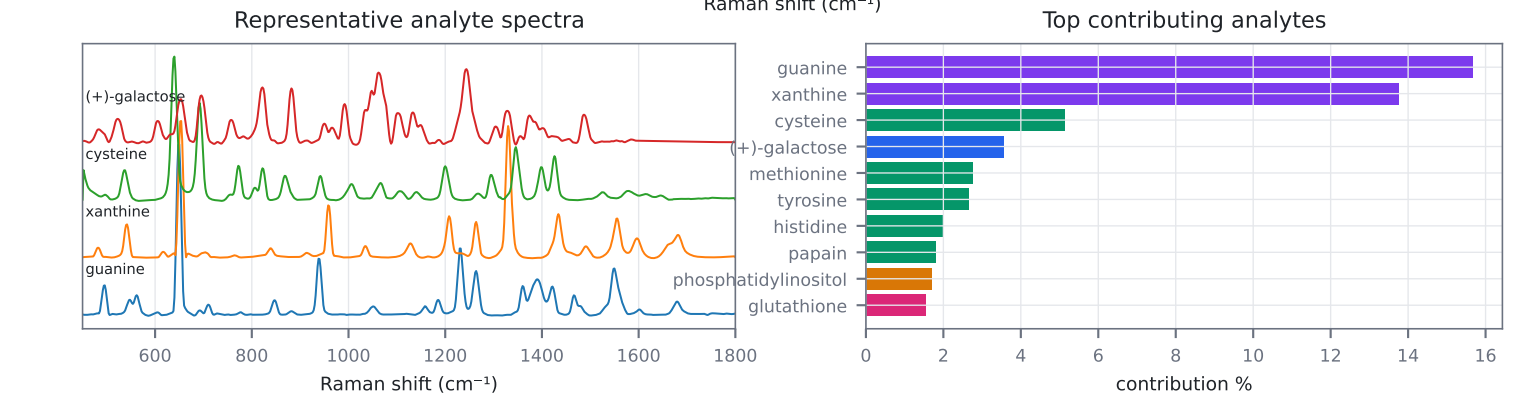
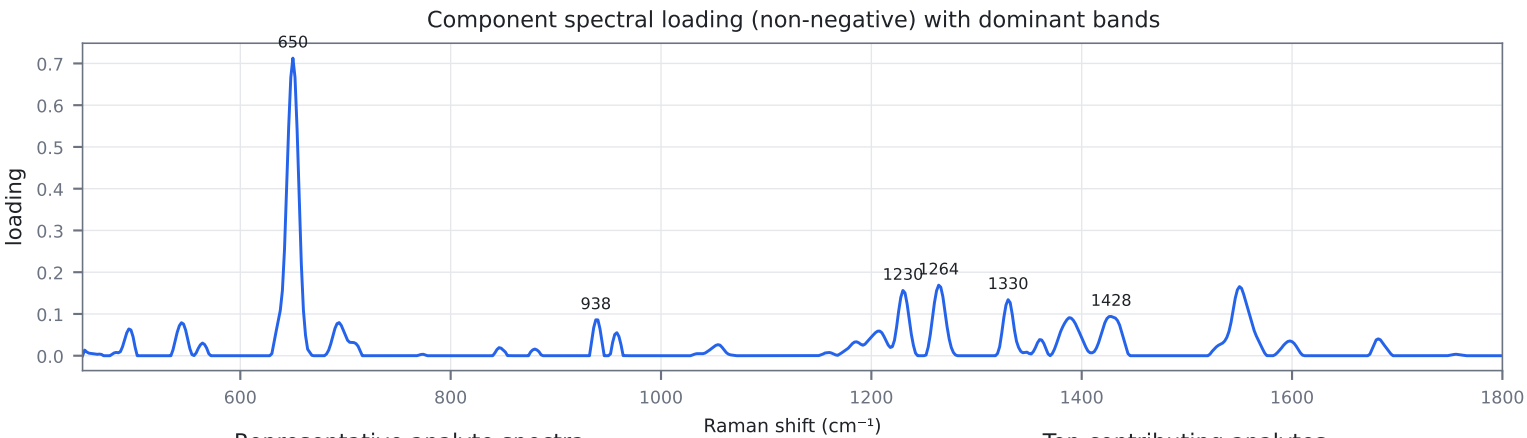
Full analyte composition — top 18 of 108 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
phosphate	7.53	cofactor/small-molecule	organic_acid	unavailable
succinic acid	6.8	cofactor/small-molecule	organic_acid	unavailable
aspartic acid	6.32	cofactor/small-molecule	organic_acid	unavailable
citrate	4.14	cofactor/small-molecule	organic_acid	unavailable
asparagine	3.97	protein/peptide	amino_acid	aliphatic/polar amino acid
amylopectin	3.29	carbohydrate	polysaccharide	unavailable
amylose	3.12	carbohydrate	polysaccharide	unavailable
xylanase	3.11	protein/peptide	protein	unavailable
valine	2.6	protein/peptide	amino_acid	aliphatic/polar amino acid
galactose	2.24	carbohydrate	saccharide	disaccharide

Component 15

tentative theme: purine · class: protein/peptide · confidence: high

variance 0.0217 stability 0.787 purity 0.303 enrichment 10.11x entropy 0.819
analytes contributing 90 dominant (≥5%) 3 sparsity 0.825 band uniqueness 0.84 median band width 13 cm⁻¹
representative analyte: guanine · secondary family: saccharide



Interpretation. The component is dominated by protein/peptide chemistry (family 'purine', 30% of its loading, 10.1x enriched over the corpus base rate). Its strongest bands lie at 650, 938, 1230, 1264, 1330 cm⁻¹. Literature corroboration: {'Proteins': 4, 'Lipids-FattyAcids': 3, 'NucleicAcids': 1, 'Saccharides-Polysaccharides': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

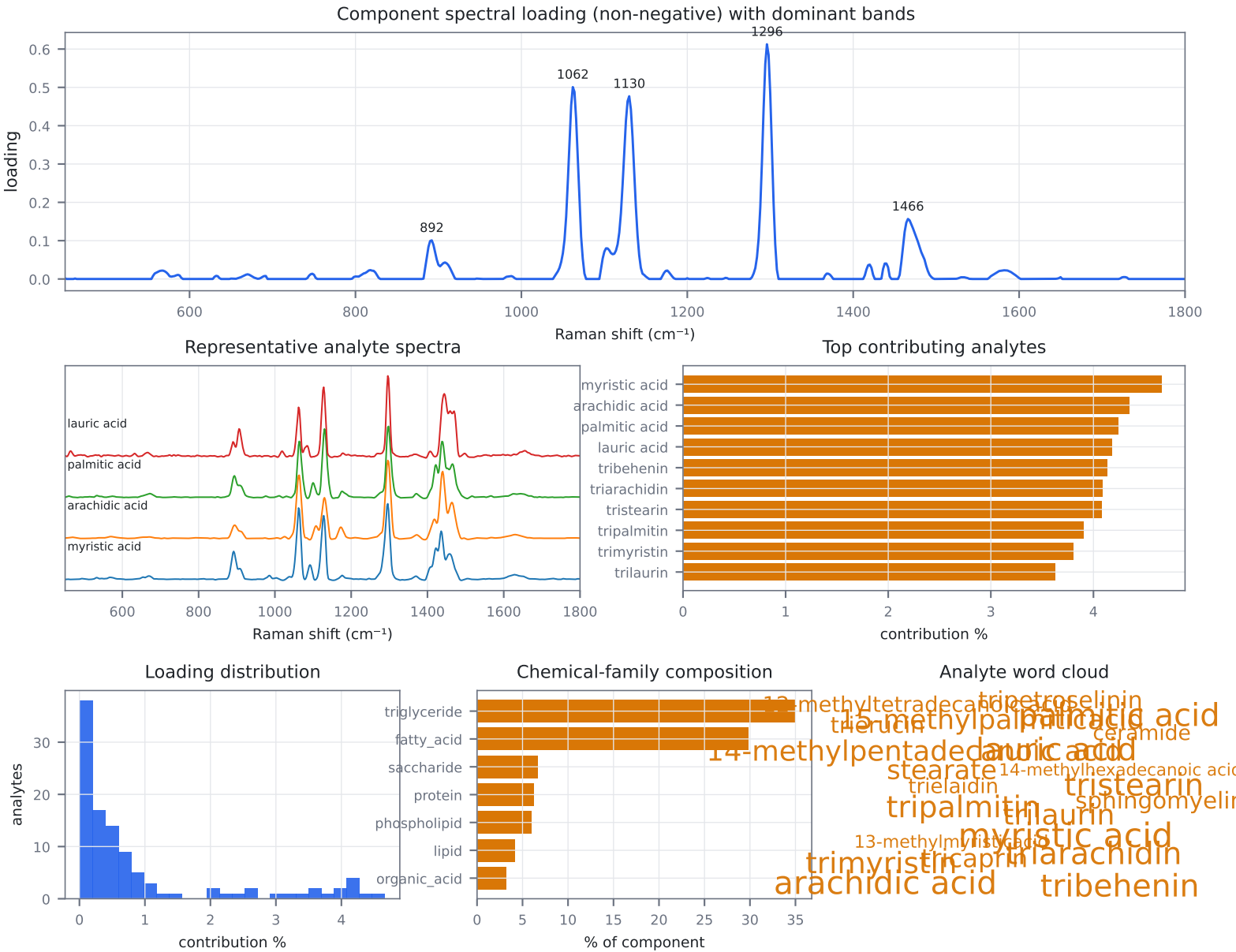
Full analyte composition — top 18 of 90 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
guanine	15.65	nucleic	purine	purine base/derivative
xanthine	13.75	nucleic	purine	purine base/derivative
cysteine	5.13	protein/peptide	amino_acid	aliphatic/polar amino acid
(+)-galactose	3.56	carbohydrate	saccharide	disaccharide
methionine	2.75	protein/peptide	amino_acid	aliphatic/polar amino acid
tyrosine	2.65	protein/peptide	amino_acid	aromatic amino acid
histidine	2.01	protein/peptide	amino_acid	aromatic amino acid
papain	1.8	protein/peptide	protein	unavailable
phosphatidylinositol	1.69	lipid	lipid	unavailable
glutathione	1.54	cofactor/small molecule	cofactor	unavailable

Component 16

tentative theme: triglyceride · class: lipid · confidence: high

variance 0.0397 stability 0.847 purity 0.350 enrichment 3.66x entropy 0.837
analytes contributing 108 dominant (≥5%) 0 sparsity 0.901 band uniqueness 0.84 median band width 13 cm⁻¹
representative analyte: myristic acid · secondary family: saccharide



Interpretation. The component is dominated by lipid chemistry (family 'triglyceride', 35% of its loading, 3.7x enriched over the corpus base rate). Its strongest bands lie at 892, 1062, 1130, 1296, 1466 cm⁻¹. Literature corroboration: {'Proteins': 3, 'Saccharides-Monosaccharides': 2, 'Lipids-FattyAcids': 1}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

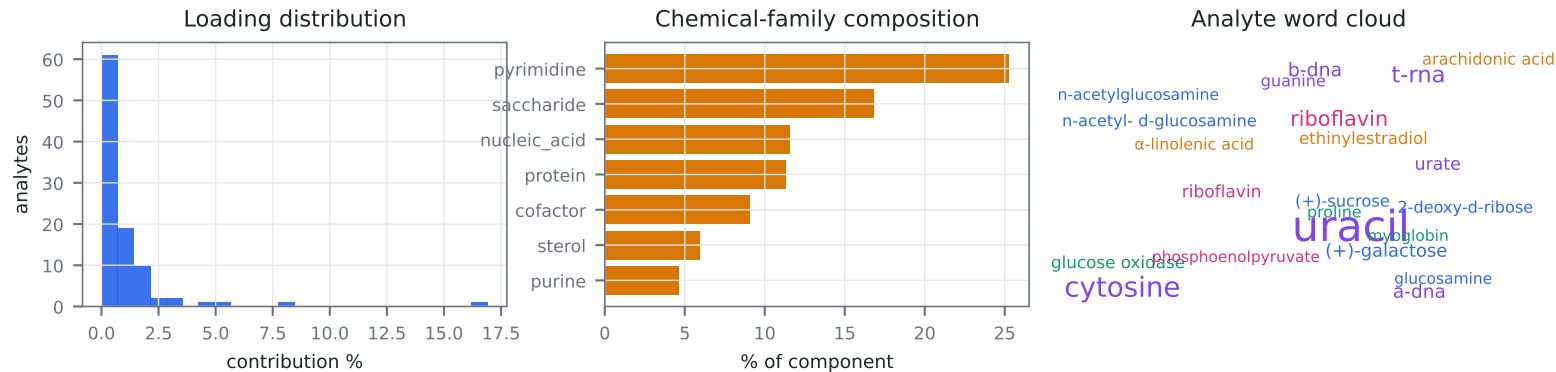
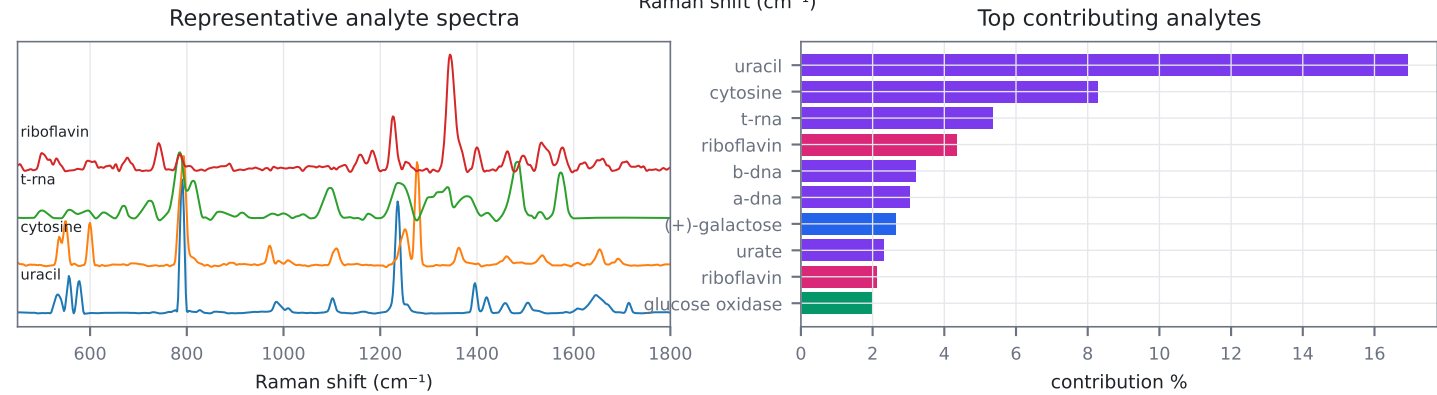
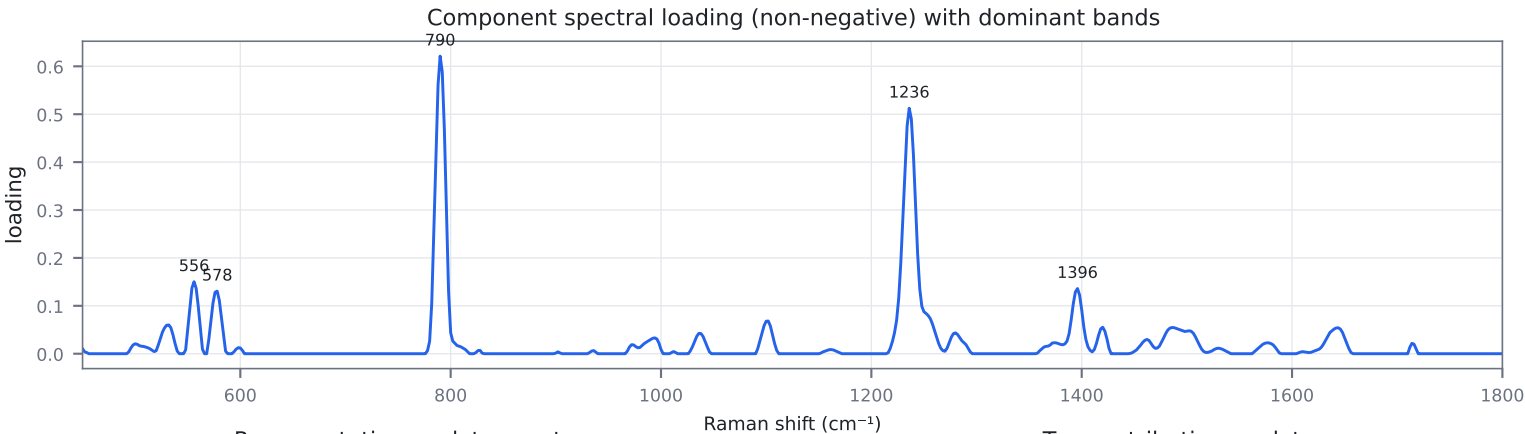
Full analyte composition — top 18 of 108 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
myristic acid	4.66	lipid	fatty_acid	saturated fatty acid
arachidic acid	4.35	lipid	fatty_acid	saturated fatty acid
palmitic acid	4.24	lipid	fatty_acid	saturated fatty acid
lauric acid	4.18	lipid	fatty_acid	saturated fatty acid
tribehenin	4.13	lipid	triglyceride	triacylglycerol
triarachidin	4.08	lipid	triglyceride	triacylglycerol
tristearin	4.08	lipid	triglyceride	triacylglycerol
tripalmitin	3.9	lipid	triglyceride	triacylglycerol
trimyristin	3.8	lipid	triglyceride	triacylglycerol
trilaurin	3.63	lipid	triglyceride	triacylglycerol

Component 17

tentative theme: pyrimidine · class: nucleic · confidence: high

variance 0.0211 stability 0.819 purity 0.253 enrichment 14.06x entropy 0.815
analytes contributing 98 dominant (≥5%) 3 sparsity 0.853 band uniqueness 0.91 median band width 11 cm⁻¹
representative analyte: uracil · secondary family: saccharide



Interpretation. The component is dominated by nucleic chemistry (family 'pyrimidine', 25% of its loading, 14.1x enriched over the corpus base rate). Its strongest bands lie at 556, 578, 790, 1236, 1396 cm⁻¹. Literature corroboration: {'Proteins': 3, 'NucleicAcids': 3, 'Lipids-FattyAcids': 2, 'Saccharides-Monosaccharides': 1}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

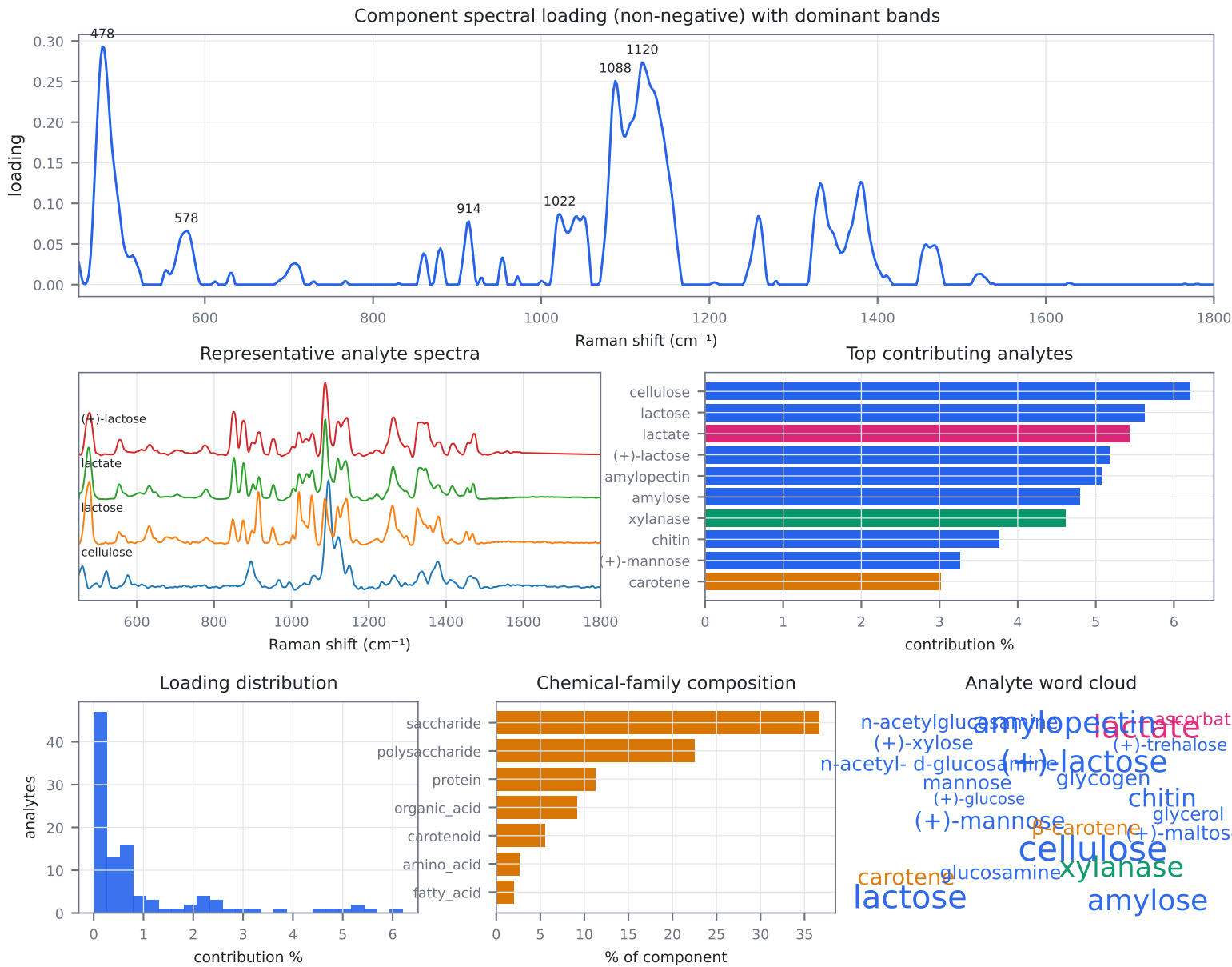
Full analyte composition — top 18 of 98 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
uracil	16.91	nucleic	pyrimidine	pyrimidine base
cytosine	8.27	nucleic	pyrimidine	pyrimidine base
t-rna	5.35	nucleic	nucleic_acid	unavailable
riboflavin	4.33	cofactor/small-molecule	cofactor	unavailable
b-dna	3.21	nucleic	nucleic_acid	unavailable
a-dna	3.02	nucleic	nucleic_acid	unavailable
(+)-galactose	2.65	carbohydrate	saccharide	disaccharide
urate	2.32	nucleic	purine	purine base/derivative
riboflavin	2.11	cofactor/small-molecule	cofactor	unavailable
glucose oxidase	1.98	protein/peptide	protein	unavailable

Component 18

tentative theme: saccharide · class: carbohydrate · confidence: high

variance 0.0370 stability 0.882 purity 0.367 enrichment 2.27x entropy 0.822
analytes contributing 105 dominant (≥5%) 5 sparsity 0.799 band uniqueness 0.85 median band width 22 cm⁻¹
representative analyte: cellulose secondary family: saccharide



Interpretation. The component is dominated by carbohydrate chemistry (family 'saccharide', 37% of its loading, 2.3x enriched over the corpus base rate). Its strongest bands lie at 478, 578, 914, 1022, 1088 cm⁻¹. Literature corroboration: {'Proteins': 5, 'Saccharides-Monosaccharides': 2, 'NucleicAcids': 2, 'Lipids-FattyAcids': 1, 'Lipids-Hormones': 1}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

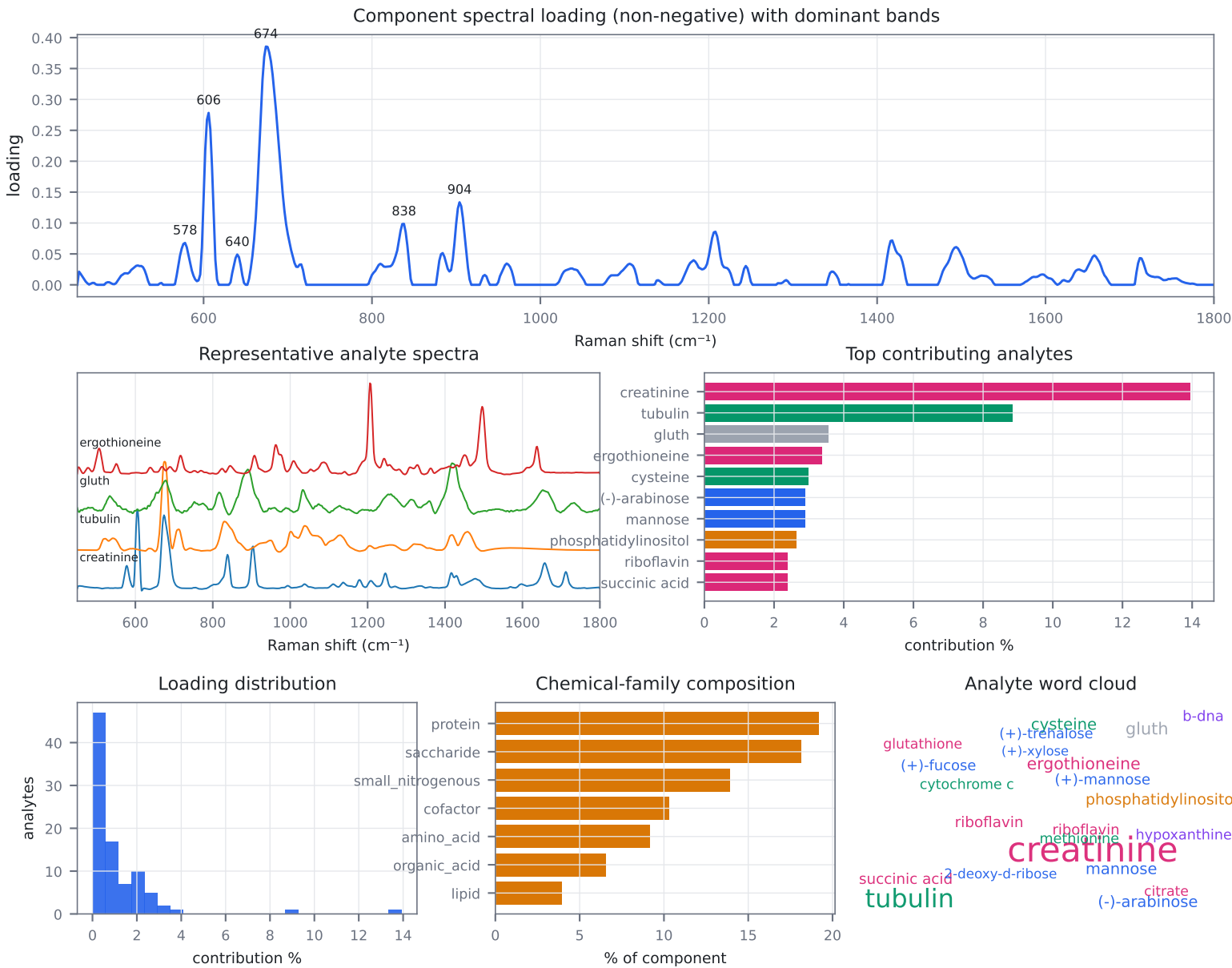
Full analyte composition — top 18 of 105 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
cellulose	6.2	carbohydrate	polysaccharide	unavailable
lactose	5.62	carbohydrate	saccharide	disaccharide
lactate	5.43	cofactor/small-molecule	organic_acid	unavailable
(+)-lactose	5.17	carbohydrate	saccharide	disaccharide
amylopectin	5.07	carbohydrate	polysaccharide	unavailable
amylose	4.79	carbohydrate	polysaccharide	unavailable
xylanase	4.61	protein/peptide	protein	unavailable
chitin	3.76	carbohydrate	polysaccharide	unavailable
(+)-mannose	3.26	carbohydrate	saccharide	monosaccharide
carotene	3.01	lipid	carotenoid	unavailable

Component 19

tentative theme: protein · class: cofactor/small-molecule · confidence: moderate

variance 0.0184 stability 0.825 purity 0.192 enrichment 1.00x entropy 0.833
analytes contributing 91 dominant (≥5%) 2 sparsity 0.760 band uniqueness 0.91 median band width 15 cm⁻¹
representative analyte: creatinine · secondary family: saccharide



Interpretation. The component is dominated by cofactor/small-molecule chemistry (family 'protein', 19% of its loading, 1.0x enriched over the corpus base rate). Its strongest bands lie at 578, 606, 640, 674, 838 cm⁻¹. Literature corroboration: {'Proteins': 7, 'Saccharides-Monosaccharides': 2, 'Lipids-FattyAcids': 2}.

Artifact assessment. Bootstrap stability is high but chemical purity is modest — most likely a genuine shared-band motif spanning several families rather than one biochemical theme.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

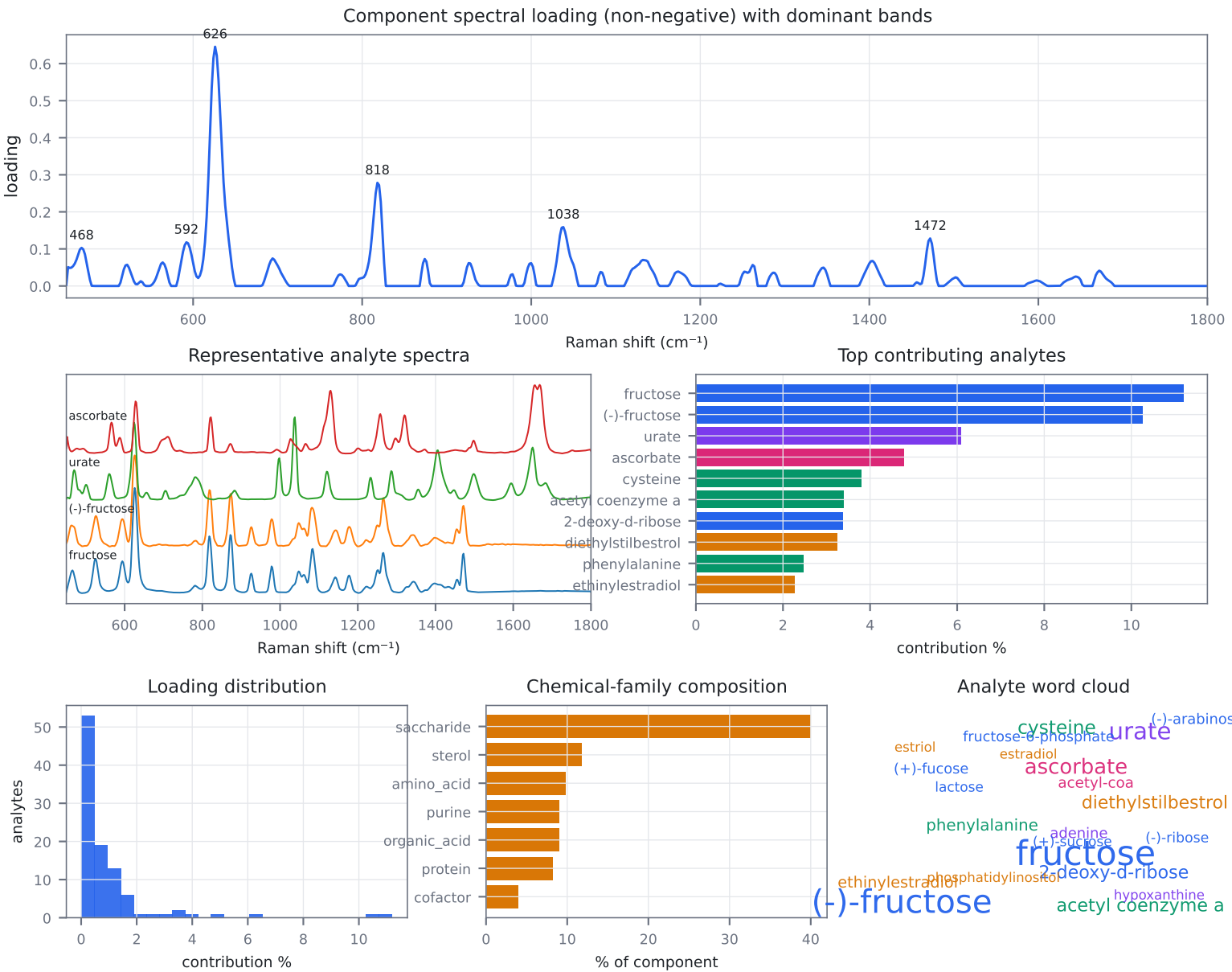
Full analyte composition — top 18 of 91 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
creatinine	13.93	cofactor/small-molecule	small_nitrogenous	unavailable
tubulin	8.83	protein/peptide	protein	unavailable
gluth	3.56	unassigned	unknown	unavailable
ergothioneine	3.37	cofactor/small-molecule	cofactor	unavailable
cysteine	2.97	protein/peptide	amino_acid	aliphatic/polar amino acid
(-)-arabinose	2.9	carbohydrate	saccharide	monosaccharide
mannose	2.9	carbohydrate	saccharide	monosaccharide
phosphatidylinositol	2.64	lipid	lipid	unavailable
riboflavin	2.38	cofactor/small-molecule	cofactor	unavailable
succinic acid	2.38	cofactor/small-molecule	organic_acid	unavailable

Component 20

tentative theme: saccharide · class: carbohydrate · confidence: high

variance 0.0212 stability 0.768 purity 0.400 enrichment 2.48x entropy 0.816
analytes contributing 101 dominant (≥5%) 3 sparsity 0.806 band uniqueness 0.93 median band width 14 cm⁻¹
representative analyte: fructose · secondary family: saccharide



Interpretation. The component is dominated by carbohydrate chemistry (family 'saccharide', 40% of its loading, 2.5x enriched over the corpus base rate). Its strongest bands lie at 468, 592, 626, 818, 1038 cm⁻¹. Literature corroboration: {'Saccharides-Monosaccharides': 3, 'Proteins': 3, 'Lipids-Hormones': 2}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

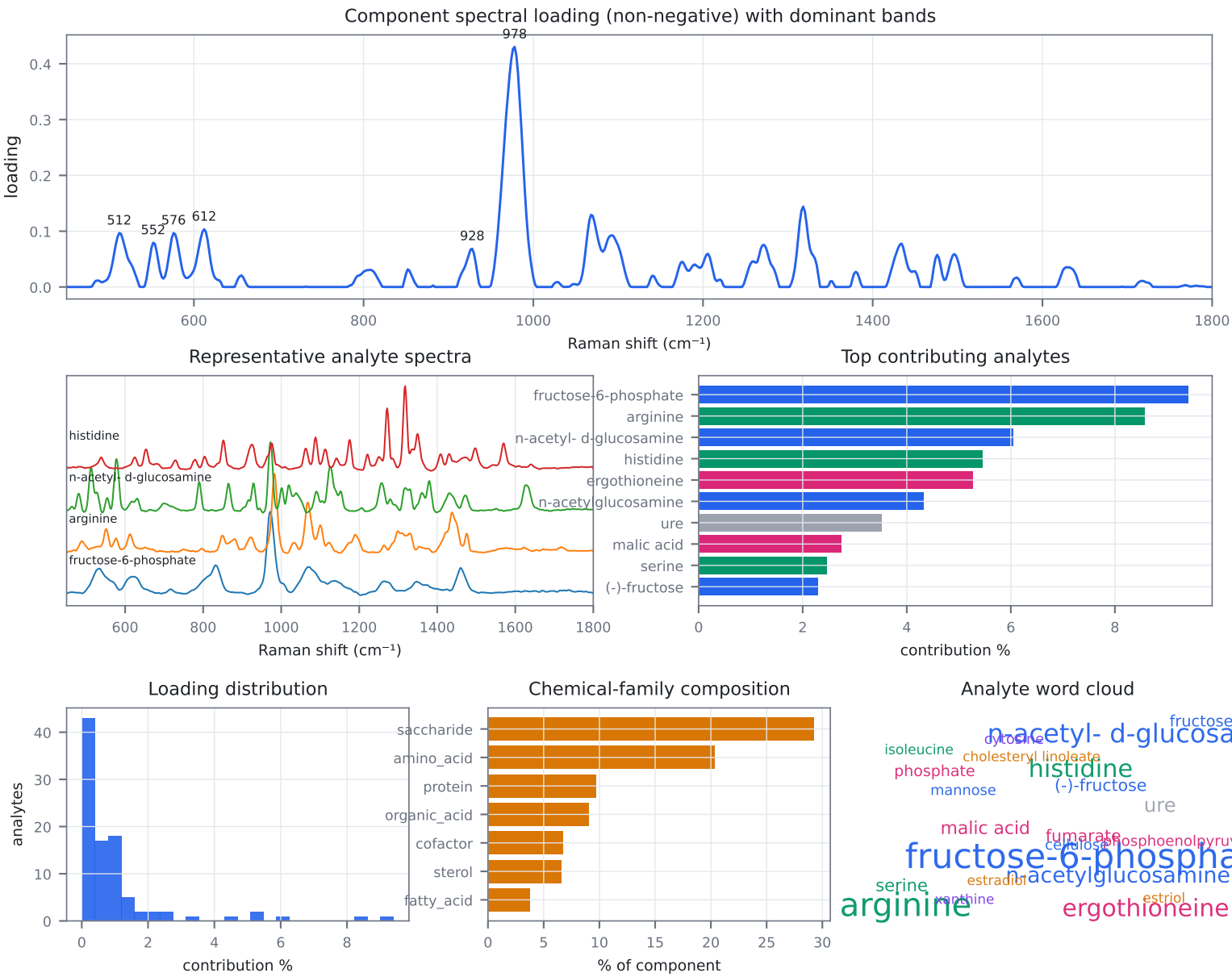
Full analyte composition — top 18 of 101 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
fructose	11.19	carbohydrate	saccharide	monosaccharide
(-)-fructose	10.26	carbohydrate	saccharide	monosaccharide
urate	6.08	nucleic	purine	purine base/derivative
ascorbate	4.78	cofactor/small-molecule	organic_acid	unavailable
cysteine	3.79	protein/peptide	amino_acid	aliphatic/polar amino acid
acetyl coenzyme a	3.4	protein/peptide	protein	unavailable
2-deoxy-d-ribose	3.38	carbohydrate	saccharide	monosaccharide
diethylstilbestrol	3.25	lipid	sterol	unavailable
phenylalanine	2.46	protein/peptide	amino_acid	aromatic amino acid
ethinylestradiol	2.26	lipid	sterol	unavailable

Component 21

tentative theme: saccharide · class: carbohydrate · confidence: moderate

variance 0.0235 stability 0.769 purity 0.293 enrichment 1.81x entropy 0.843
analytes contributing 96 dominant (≥5%) 5 sparsity 0.763 band uniqueness 0.88 median band width 17 cm⁻¹
representative analyte: fructose-6-phosphate secondary family: saccharide



Interpretation. The component is dominated by carbohydrate chemistry (family 'saccharide', 29% of its loading, 1.8x enriched over the corpus base rate). Its strongest bands lie at 512, 552, 576, 612, 928 cm⁻¹. Literature corroboration: {'Lipids-FattyAcids': 5, 'Proteins': 5, 'AminoAcids': 2, 'NucleicAcids': 2, 'Saccharides-Monosaccharides': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

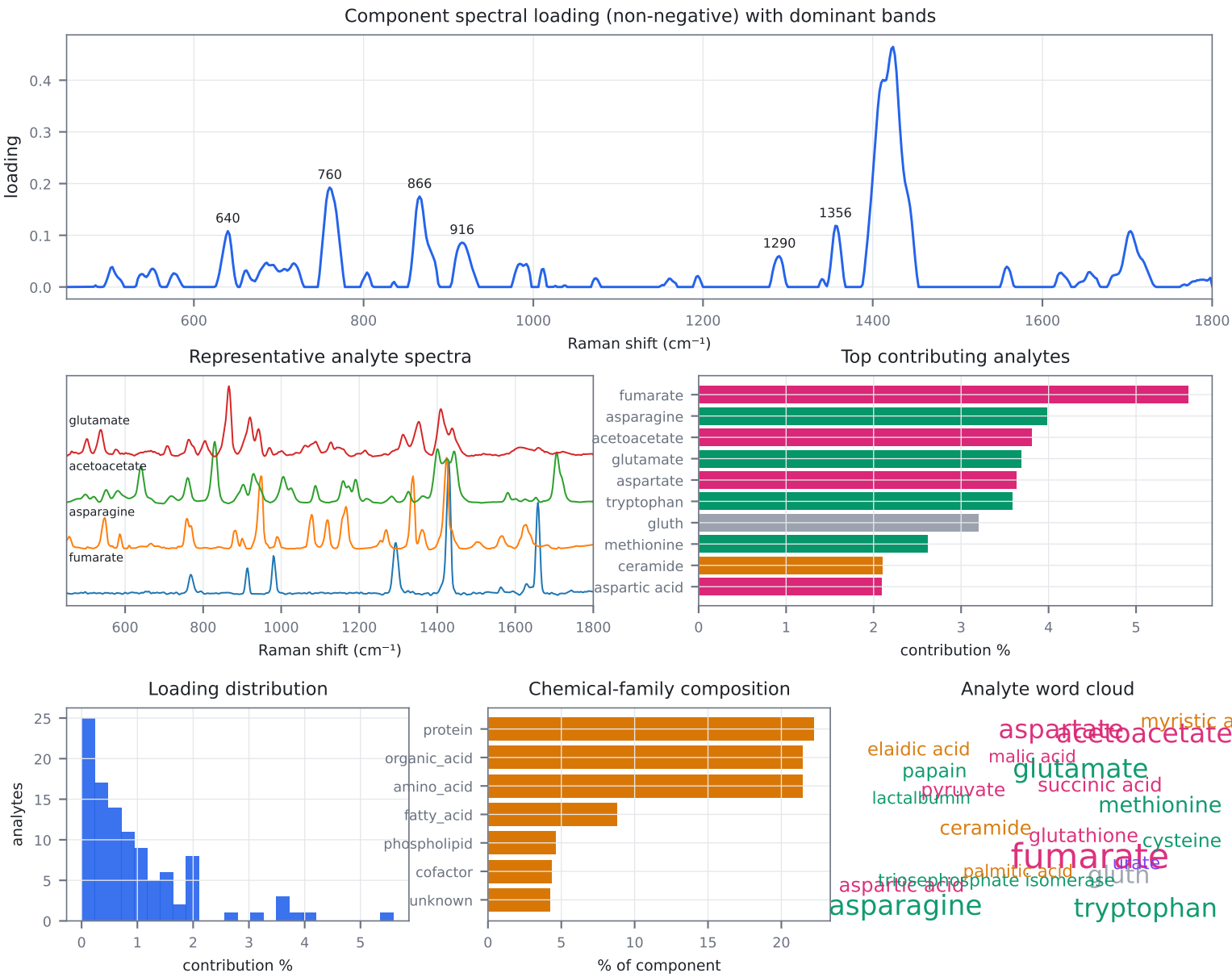
Full analyte composition — top 18 of 96 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
fructose-6-phosphate	9.4	carbohydrate	saccharide	monosaccharide
arginine	8.57	protein/peptide	amino_acid	aliphatic/polar amino acid
n-acetyl- d-glucosamine	6.04	carbohydrate	saccharide	monosaccharide
histidine	5.46	protein/peptide	amino_acid	aromatic amino acid
ergothioneine	5.27	cofactor/small-molecule	cofactor	unavailable
n-acetylglucosamine	4.33	carbohydrate	saccharide	monosaccharide
ure	3.51	unassigned	unknown	unavailable
malic acid	2.73	cofactor/small-molecule	organic_acid	unavailable
serine	2.46	protein/peptide	amino_acid	aliphatic/polar amino acid
(-) fructose	2.28	carbohydrate	saccharide	monosaccharide

Component 22

tentative theme: protein · class: protein/peptide · confidence: moderate

variance 0.0289 stability 0.735 purity 0.222 enrichment 1.16x entropy 0.897
analytes contributing 105 dominant (≥5%) 1 sparsity 0.814 band uniqueness 0.89 median band width 18 cm⁻¹
representative analyte: fumarate · secondary family: amino_acid



Interpretation. The component is dominated by protein/peptide chemistry (family 'protein', 22% of its loading, 1.2x enriched over the corpus base rate). Its strongest bands lie at 640, 760, 866, 916, 1290 cm⁻¹. Literature corroboration: {'Lipids-FattyAcids': 6, 'NucleicAcids': 2, 'Proteins': 2, 'AminoAcids': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

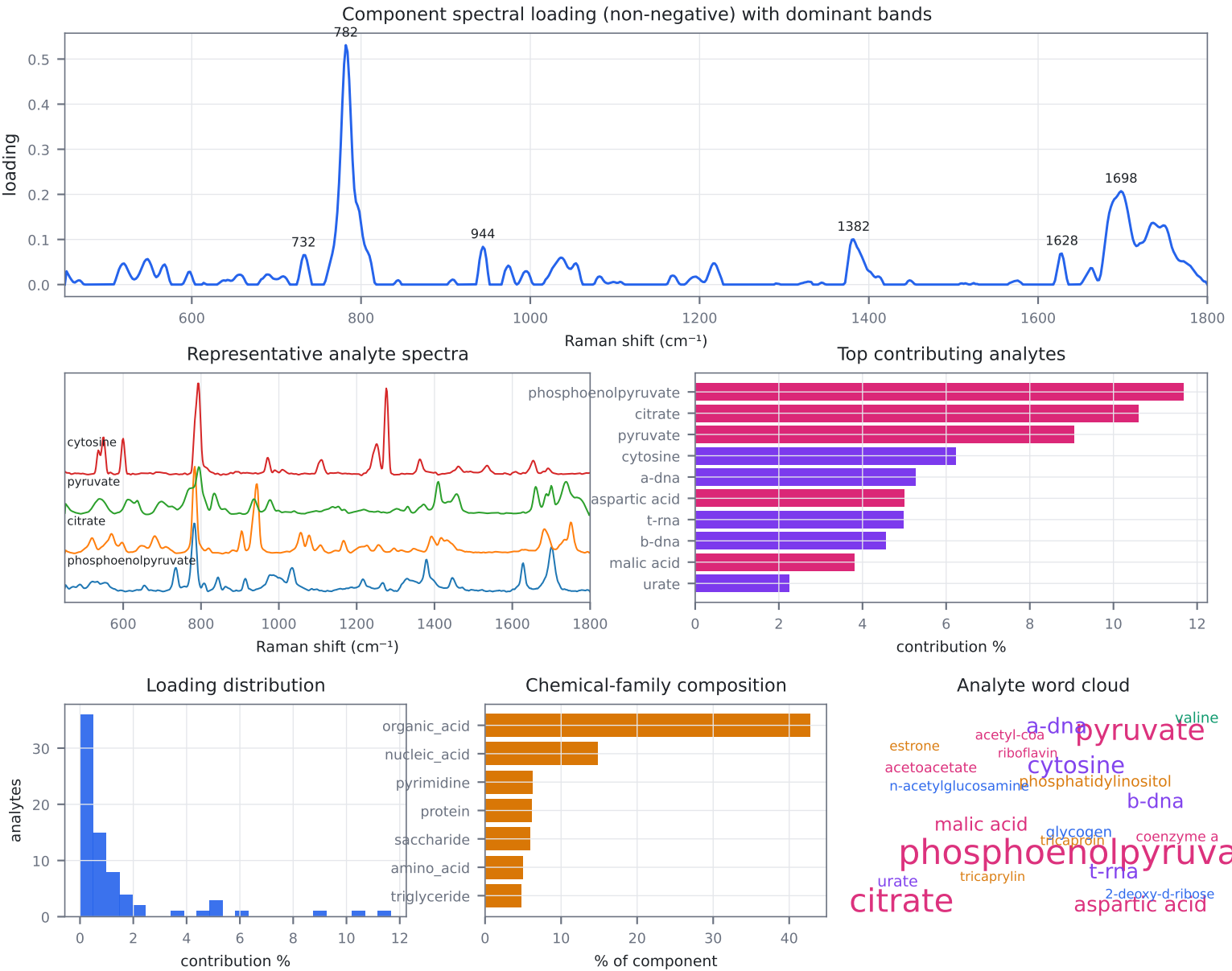
Full analyte composition — top 18 of 105 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
fumarate	5.59	cofactor/small-molecule	organic_acid	unavailable
asparagine	3.98	protein/peptide	amino_acid	aliphatic/polar amino acid
acetoacetate	3.8	cofactor/small-molecule	organic_acid	unavailable
glutamate	3.69	protein/peptide	amino_acid	aliphatic/polar amino acid
aspartate	3.63	cofactor/small-molecule	organic_acid	unavailable
tryptophan	3.58	protein/peptide	amino_acid	aromatic amino acid
gluth	3.2	unassigned	unknown	unavailable
methionine	2.61	protein/peptide	amino_acid	aliphatic/polar amino acid
ceramide	2.1	lipid	phospholipid	unavailable
aspartic acid	2.09	cofactor/small-molecule	organic_acid	unavailable

Component 23

tentative theme: organic_acid · class: cofactor/small-molecule · confidence: high

variance 0.0196 stability 0.781 purity 0.428 enrichment 5.95x entropy 0.793
analytes contributing 74 dominant (≥5%) 5 sparsity 0.787 band uniqueness 0.93 median band width 13 cm⁻¹
representative analyte: phosphoenolpyruvate · secondary family: triglyceride



Interpretation. The component is dominated by cofactor/small-molecule chemistry (family 'organic_acid', 43% of its loading, 6.0x enriched over the corpus base rate). Its strongest bands lie at 732, 782, 944, 1382, 1628 cm⁻¹. Literature corroboration: {'NucleicAcids': 3, 'Proteins': 3, 'Lipids-FattyAcids': 1, 'Saccharides-Polysaccharides': 1, 'Lipids-Hormones': 1}.

Artifact assessment. Bootstrap stability is below 0.80: this component may be partly a fitting artifact and should not anchor an ontology entry on its own.

This is a mathematical component. The theme is a tentative post-hoc interpretation, never a molecular assignment.

Full analyte composition — top 18 of 74 contributing analytes (molecular weight and biochemical role: unavailable in this corpus)

analyte	%	class	family	subfamily
phosphoenolpyruvate	11.66	cofactor/small-molecule	organic_acid	unavailable
citrate	10.59	cofactor/small-molecule	organic_acid	unavailable
pyruvate	9.05	cofactor/small-molecule	organic_acid	unavailable
cytosine	6.22	nucleic	pyrimidine	pyrimidine base
a-dna	5.27	nucleic	nucleic_acid	unavailable
aspartic acid	5.0	cofactor/small-molecule	organic_acid	unavailable
t-rna	4.97	nucleic	nucleic_acid	unavailable
b-dna	4.56	nucleic	nucleic_acid	unavailable
malic acid	3.79	cofactor/small-molecule	organic_acid	unavailable
urate	2.25	nucleic	purine	purine base/derivative

Chemical coherence

P4 — chemical coherence ranking (most → least coherent)

comp	purity	enrich	entropy	MI	families	specSim	dominant family
2	0.803	4.19	0.343	0.423	7	0.937	protein
12	0.544	3.37	0.613	0.401	11	0.529	saccharide
10	0.521	3.22	0.644	0.344	12	0.535	saccharide
4	0.429	2.65	0.747	0.217	14	0.481	saccharide
23	0.428	5.95	0.768	0.325	12	0.447	organic_acid
20	0.400	2.48	0.731	0.312	11	0.348	saccharide
11	0.371	3.64	0.723	0.356	13	0.495	amino_acid
18	0.367	2.27	0.713	0.324	12	0.535	saccharide
8	0.360	1.88	0.742	0.336	12	0.426	protein
16	0.350	3.66	0.693	0.297	11	0.913	triglyceride
6	0.335	1.75	0.751	0.443	11	0.469	protein
0	0.333	1.74	0.675	0.515	10	0.497	protein
1	0.322	3.36	0.659	0.712	9	0.841	triglyceride
15	0.303	10.11	0.734	0.361	10	0.397	purine
7	0.293	1.53	0.738	0.333	13	0.748	protein
21	0.293	1.81	0.807	0.285	14	0.336	saccharide
5	0.281	2.76	0.706	0.392	10	0.417	amino_acid
14	0.280	3.89	0.786	0.290	12	0.438	organic_acid
17	0.253	14.06	0.779	0.362	11	0.331	pyrimidine
9	0.242	2.38	0.813	0.231	12	0.474	amino_acid
22	0.222	1.16	0.784	0.373	13	0.439	protein
3	0.217	4.03	0.819	0.416	11	0.458	sterol
19	0.192	1.00	0.834	0.314	14	0.355	protein
13	0.187	10.39	0.883	0.301	15	0.438	pyrimidine

Average molecular similarity is reported as UNAVAILABLE: this corpus contains no structures, SMILES or fingerprints, so a structural similarity cannot be computed without inventing metadata.

True biochemical themes vs mathematical mixtures

Components with purity ≥ 0.40 and stability ≥ 0.80 behave as genuine biochemical themes (1 of 24). The remainder are better described as stable mathematical mixtures that capture shared Raman motifs — most often the acyl-chain and C-H bands common to all lipid subclasses, and the C-O/C-C ring modes common to carbohydrates.

Higher-order grouping

P7 — grouping study (recommended k=14, provisional)

k	silhouette	bootstrap	chem	interp	composite
6	0.059	0.884	0.259	0.00	0.000
8	0.078	0.900	0.260	0.00	0.162
10	0.079	0.938	0.281	0.10	0.501
12	0.094	0.960	0.308	0.17	0.872
14	0.095	0.969	0.308	0.21	0.953
16	0.083	0.976	0.312	0.19	0.908

Silhouette is low (≤ 0.10) at every k: the components do not form geometrically well-separated clusters.

Bootstrap reproducibility is high because the same weak structure is recovered consistently —
reproducible does not mean well-separated. Grouping is therefore justified as an interpretive overlay only.

Group composition at k=14

grp	n	share	dominant family (fraction)
1	2	0.213	triglyceride (0.31)
11	4	0.180	saccharide (0.40)
2	2	0.108	protein (0.57)
5	2	0.097	organic_acid (0.26)
4	3	0.073	protein (0.28)
12	2	0.065	saccharide (0.41)
3	2	0.054	organic_acid (0.20)
14	1	0.039	protein (0.34)
7	1	0.036	protein (0.19)
13	1	0.034	saccharide (0.29)
6	1	0.031	amino_acid (0.24)
8	1	0.026	sterol (0.22)
9	1	0.025	purine (0.30)
10	1	0.019	amino_acid (0.28)

Provisional ontology v0.1

P10 — provisional ontology v0.1 (NOT frozen)

HIGH CONFIDENCE — 6 components

c2	protein	purity 0.80	enrich	4.2x	stab 0.96
c4	saccharide	purity 0.43	enrich	2.6x	stab 0.77
c10	saccharide	purity 0.52	enrich	3.2x	stab 0.65
c12	saccharide	purity 0.54	enrich	3.4x	stab 0.79
c20	saccharide	purity 0.40	enrich	2.5x	stab 0.77
c23	organic_acid	purity 0.43	enrich	6.0x	stab 0.78

MODERATE CONFIDENCE — 10 components

c0	protein	purity 0.33	enrich	1.7x	stab 0.81
c1	triglyceride	purity 0.32	enrich	3.4x	stab 0.95
c6	protein	purity 0.34	enrich	1.8x	stab 0.87
c8	protein	purity 0.36	enrich	1.9x	stab 0.93
c11	amino_acid	purity 0.37	enrich	3.6x	stab 0.89
c14	organic_acid	purity 0.28	enrich	3.9x	stab 0.77
c15	purine	purity 0.30	enrich	10.1x	stab 0.79
c16	triglyceride	purity 0.35	enrich	3.7x	stab 0.85
c17	pyrimidine	purity 0.25	enrich	14.1x	stab 0.82
c18	saccharide	purity 0.37	enrich	2.3x	stab 0.88

LOW CONFIDENCE — 6 components

c3	sterol	purity 0.22	enrich	4.0x	stab 0.82
c5	amino_acid	purity 0.28	enrich	2.8x	stab 0.74
c7	protein	purity 0.29	enrich	1.5x	stab 0.95
c9	amino_acid	purity 0.24	enrich	2.4x	stab 0.71
c21	saccharide	purity 0.29	enrich	1.8x	stab 0.77
c22	protein	purity 0.22	enrich	1.2x	stab 0.73

UNKNOWN — 2 components

c13	pyrimidine	purity 0.19	enrich	10.4x	stab 0.66
c19	protein	purity 0.19	enrich	1.0x	stab 0.82

Caveat: Themes are tentative interpretations of latent components, not molecular assignments.

Caveat: Mean component class purity is only 0.35: most components are chemically enriched rather than chemically pure.

Caveat: The atlas resolves molecular CLASS reliably; species within a class are frequently not resolved (see P12).

BSV design study

P11 — BSV design study · recommendation: C_hierarchical

A_24_canonical: Use all 24 latent components directly as BSV coordinates.

interpretability : low-moderate – 11/24 components are only moderately interpretable and mean class purity is 0.35
stability : high – mean bootstrap component stability 0.812
extensibility : high – new analytes project without changing the basis
clinical usability: poor – 24 weakly-named axes are not communicable
verdict : retain as the CANONICAL numerical layer, not as the reported BSV

B_compressed: Replace the 24 components with ~14 higher-order coordinates.

interpretability : moderate – groups gain names but lose component detail
stability : grouping bootstrap reproducibility 0.969 but silhouette only 0.095
extensibility : moderate – a new chemistry may not fit existing groups
clinical usability: good
verdict : not sufficient alone – discards resolution the atlas actually has

C_hierarchical: 24 latent components (canonical, frozen) → higher-order biochemical themes (reported) → optional visual summary.

interpretability : good – evidence is traceable from a reported theme back to components, bands and reference analytes
stability : inherits component stability; group layer is re-derivable and versionable
extensibility : high – the numerical layer never changes when the ontology is revised
clinical usability: good
verdict : RECOMMENDED

Rationale

- The 24-component layer is stable (mean bootstrap stability 0.812) and is the level at which the atlas is actually reproducible, so it should remain canonical and frozen.
- It is NOT the level at which the atlas is interpretable: mean class purity is 0.35 and only 13/24 components reach high interpretive confidence.
- Grouping is reproducible (bootstrap 0.969) but geometrically weak (silhouette 0.095), so the group layer should be treated as an interpretive overlay that can be revised, never as the frozen numerical basis.
- A hierarchy keeps both properties: numbers from the components, meaning from the themes.

The BSV is NOT defined or frozen by this audit; only the architecture is recommended.

MSS readiness

P12 — MSS readiness

Median signature uniqueness 0.053 across 167 analytes; against genuinely different chemistry it rises to 0.148.

nearest-neighbour relation: {'family': 66, 'homolog': 33, 'distinct': 32, 'same_class': 26, 'duplicate': 10}
low-uniqueness (<0.15) relations: {'family': 57, 'homolog': 28, 'same_class': 23, 'distinct': 16, 'duplicate': 10}
88% of low-uniqueness cases are duplicate entries, homologous series or same-class chemistry. The saturated triacylglycerols (trilaurin/trimyrustin/tripalmitin/tristearin/triarachidin/tribehenin) are mutually indistinguishable because in the 450–1800 cm⁻¹ fingerprint region they differ only in CH₂ count; estradiol and estriol are likewise near-identical. This is correct spectroscopy.

Verdict

MSS can be frozen at CLASS/THEME level with explicit confusable-group annotations. It cannot yet be frozen at molecular-species level. Species-level resolution would require spectral range beyond the fingerprint region (the 2800–3000 cm⁻¹ C–H stretch region separates acyl chain lengths) or an orthogonal measurement — not a change to this atlas.

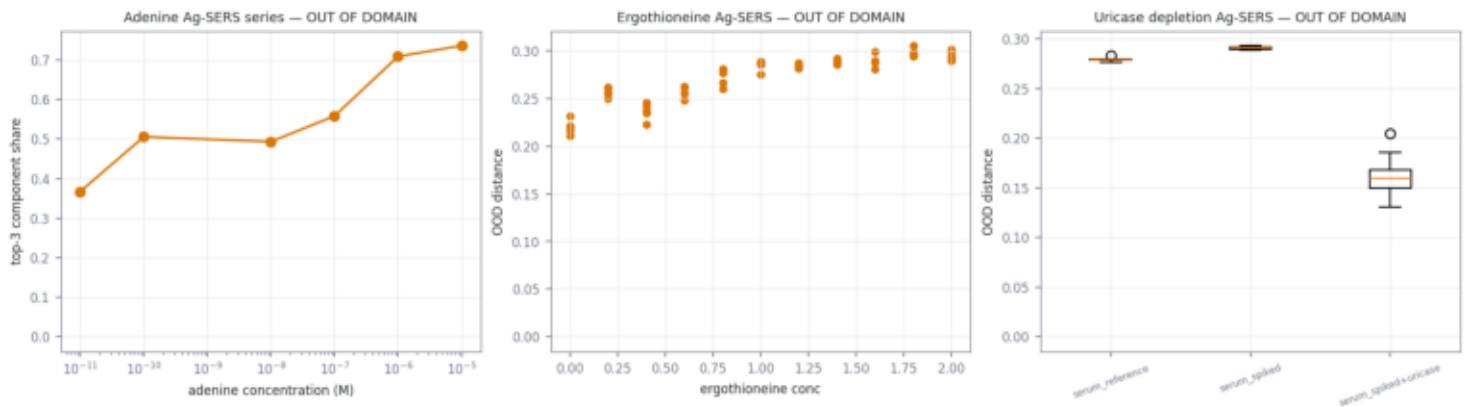
Genuine cross-chemistry confusions

xylanase	~ amylopectin	cos=0.997
(+)-lactose	~ lactate	cos=0.993
lactate	~ (+)-lactose	cos=0.993
lactose	~ lactate	cos=0.980
creatinine	~ tubulin	cos=0.965
tubulin	~ creatinine	cos=0.965
ure	~ urea	cos=0.947
urea	~ ure	cos=0.947

Appendix — out-of-domain stress test

NOT VALIDATION. Ag-SERS spectra projected into a Raman atlas.

P13 — Ag-SERS projected into a RAMAN atlas: stress test only, not validation



Three Ag-SERS calibration sets were projected into the frozen Raman atlas purely to observe its behaviour off-domain: the adenine concentration series (6 spectra), the ergothioneine calibration (55), and the uricase depletion experiment (20). Median out-of-domain distance is 0.279, materially higher than for in-domain Raman references, exactly as expected for a different observation physics. These projections were NOT used to modify, tune or judge the atlas, and no dose-response claim is made from them: prior work established that Ag-SERS spectra in this corpus are background-dominated, so any apparent trajectory here is not evidence of biochemical response.

Final scientific assessment

1. What chemistry has the atlas learned?

Lipid chemistry dominates (triacylglycerols, fatty acids, sterols cohere with $p < 0.05$ against a random-analyte null), followed by carbohydrates (mono- and polysaccharides) and proteins. Nucleic-acid bases appear as strongly ENRICHED but small components (purine enrichment 10x, pyrimidine 14x). Amino acids, cofactors and organic acids do NOT form coherent groups at analyte level.

4. Should 24 remain canonical?

Yes as the numerical layer. The components are the level at which the atlas is stable (mean bootstrap stability 0.812); no evidence supports changing k , and k was frozen for this audit in any case.

5. Should components be grouped?

Yes, but only as an interpretive overlay. Grouping is highly reproducible (0.969) yet geometrically weak (silhouette 0.095) — the components do not form well-separated natural clusters.

6. What grouping is most defensible?

$k=14$ by the composite of reproducibility, chemical coherence, silhouette and interpretable-group fraction. Ranking is reported in full; the margin over neighbouring k is small, so the group count should be treated as provisional.

7. Should the BSV be 24-D or compressed?

Hierarchical (Option C): 24 canonical latent + reported themes.

8. Is MSS mature enough?

Not at molecular-species level. Median signature uniqueness is 0.053; however 88% of low-uniqueness cases are duplicates, homologous series or same-class chemistry, which is correct spectroscopy rather than atlas failure. MSS can be frozen at CLASS/THEME level with documented confusable groups; only 16 genuinely cross-chemistry confusions remain.

10. What should the next GAIRA phase be?

Define the hierarchical BSV v0.1 on the frozen 24-component layer with a versioned, revisable theme overlay, and publish MSS at class level with explicit confusable-group annotations. Do not expand k , do not retrain, and do not admit Ag-SERS.

2. Biologically strongest components

c2 protein	purity 0.80 enrich	4.2x stab 0.96
c12 saccharide	purity 0.54 enrich	3.4x stab 0.79
c10 saccharide	purity 0.52 enrich	3.2x stab 0.65
c23 organic_acid	purity 0.43 enrich	6.0x stab 0.78
c11 amino_acid	purity 0.37 enrich	3.6x stab 0.89

3. Weakest components

c19 protein	purity 0.19 enrich	1.0x stab 0.83
c22 protein	purity 0.22 enrich	1.2x stab 0.74
c9 amino_acid	purity 0.24 enrich	2.4x stab 0.71
c13 pyrimidine	purity 0.19 enrich	10.4x stab 0.66
c21 saccharide	purity 0.29 enrich	1.8x stab 0.77

9. Ontology confidence

```
{'high_confidence': 6, 'moderate_confidence': 10, 'low_confidence': 6, 'unknown': 2}
```